

DESIGN & DEVELOPMENT OF A VIBRATION ENERGY HARVESTING SYSTEM FOR A STEALTH VEHICLE TRACKING UNIT

210286R	Kelaart D.J.
210317P	Kumarasinghe K.A.T.K
210377X	Manukulasooriya C.A.

Advised by

Advisor name and affiliation.
Dr. Mahesh Hordagoda
Department of Mechanical Engineering
University of Moratuwa

Report submitted in partial fulfilment of the requirements for the degree of Bachelor of
Science in Engineering

Department of Mechanical Engineering

University of Moratuwa
Sri Lanka

May 2026

Declaration page of the candidate and supervisor

We declare that this is our own work and this report does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Also, we hereby grant to University of Moratuwa the non-exclusive right to reproduce and distribute our report, in whole or in part in print, electronic or other medium.

Signature: Decon. Date: 17/05/2026

Signature: Thingf.. Date: 17/05/2026

Signature: [Signature] Date: 17/05/2026

The supervisor/s should certify the report with the following declaration.

The above candidates have carried out the research included in this research under my supervision.

Signature of the supervisor: [Signature] Date 17/05/2026

Acknowledgements

We would like to express our sincere gratitude to our project supervisor, Dr. Mahesh Hordagoda, for his continuous guidance, encouragement, and valuable technical advice throughout this project. His support and constructive feedback greatly helped us in completing this research successfully.

We also wish to extend our sincere thanks to Dr. Logeeshan for his valuable support, technical discussions, and guidance related to the electrical and electronic aspects of the project.

Our appreciation is also extended to the academic staff members of the Department of Mechanical Engineering and the Department of Electrical Engineering, University of Moratuwa, for sharing their knowledge and providing valuable suggestions during the progress of this project.

We are especially thankful to the non-academic staff members of the Material Engineering laboratories and other university laboratories for providing access to laboratory facilities, equipment, and measurement support required for fabrication, testing, and experimental work.

In addition, we would like to thank all the non-academic staff members of the University of Moratuwa for their assistance and cooperation throughout the duration of this project.

Finally, we would like to express our heartfelt gratitude to our parents, friends, and colleagues for their continuous encouragement, motivation, and support throughout the completion of this project.

Abstract

Vibration energy harvesting is a promising method for powering low-power electronic devices in applications where frequent battery replacement or manual recharging is impractical. This project focuses on the design and development of an electromagnetic vibration energy harvester to recharge the battery of a stealth GPS tracking unit intended for vehicle-mounted military tracking applications. The system is designed to utilize vertical vibration from the vehicle lower control arm as the mechanical excitation source.

The methodology involved literature review, transduction mechanism selection, theoretical modelling, numerical simulation, experimental vibration measurement, prototype fabrication, and performance testing. A Toyota Raize lower control arm was instrumented using an accelerometer-based data acquisition system, and vibration data were collected under different road profiles. The vertical acceleration component was processed using MATLAB to identify dominant vibration frequencies suitable for resonance tuning. Based on these frequencies, the harvester parameters were determined and iteratively refined using analytical calculations and COMSOL Multiphysics simulations to estimate magnetic flux variation, induced voltage, and expected electrical output.

A physical electromagnetic harvester prototype was then fabricated and experimentally tested to evaluate its voltage generation and frequency response. The results confirmed that output performance is strongly dependent on resonance matching, relative motion between the magnet and coil, and electrical loading. The study demonstrates that vehicle suspension vibrations can provide a useful supplementary energy source for extending GPS tracker battery life. However, stable long-term operation requires further optimization of the mechanical tuning, coil-magnet configuration, rectification circuit, and energy management system. Future work is recommended to improve power conversion efficiency, integrate the harvester with the battery charging circuit, and validate the complete system under real vehicle operating conditions.

Keywords: Electromagnetic energy harvesting, vehicle vibration, GPS tracking unit, resonance tuning, battery charging.

Table of Contents

Declaration page of the candidate and supervisor	i
Acknowledgements.....	ii
Abstract	iii
List of Figures	vii
List of Tables	ix
List of abbreviations	x
List of Appendices	xi
1 Introduction.....	1
1.1 Background	1
1.2 Problem Statement	1
1.3 Aim	2
1.4 Objectives	2
1.5 Scope of the Project	2
1.6 Current Project Progress	3
2 Literature Review.....	4
2.1 Piezoelectric Energy Harvesting Mechanism	4
2.2 Electromagnetic Energy Harvesting Mechanism.....	7
2.3 Comparison of Transduction Mechanisms and Selection Justification	8
2.4 Vehicle Vibration Measurement and Data Analysis for Energy Harvesting	9
2.5 GPS Tracker Selection and Energy Demand	10
2.5.1 Types of GPS Trackers and Suitability for VEH Applications	11
2.5.2 Selection of a Suitable Wireless Tracker	11
2.6 Research Gap	12
3 Methodology	13
3.1 Introduction.....	13
3.2 Calculation power requirement of the GPS tracking unit.....	13
3.3 Selection of Transduction Mechanism.....	15
3.4 Analytical Model Development.....	15
3.4.1 Introduction.....	15
3.4.2 Mechanical Model	16
3.4.3 Natural Frequency and Spring Stiffness Selection	17
3.4.4 Damping ratio and Mechanical Quality Factor.....	18
3.4.5 Frequency Response of the Mechanical System.....	18
3.4.6 Electromagnetic Model.....	19
3.4.7 Dimensionless Parameters	21

3.4.8	Normalized Governing Equations at Non-resonant Excitation	22
3.4.9	Governing Equation for Power Output at Resonance.....	25
3.4.10	Design Implications from Analytical Model	26
3.5	Experimental Vibration Data Acquisition and Signal Processing	26
3.6	PSD Analysis using Welch's Method for Road Test Vibration Data.....	29
3.6.1	Gravity component removal from the data	29
3.6.2	Frequency-Domain Representation	30
3.6.3	Manual Implementation of Welch's Method in MATLAB.....	31
3.6.4	Road PSD-Based Power Estimation	32
3.6.5	Interpretation of PSD Results	34
3.7	Numerical and Parametric Analysis.....	37
3.7.1	Effect of excitation frequency.....	38
3.7.2	Effect of Squared Electromechanical Coupling Coefficient.....	38
3.7.3	Effect of Mechanical Quality Factor.....	39
3.7.4	Effect of Resistive Loss Coefficient	39
3.8	Design and Parameter Selection	40
3.8.1	Introduction.....	40
3.8.2	Design Requirements	40
3.8.3	Conceptual Design	41
3.8.4	Main Components.....	44
3.8.5	Mathematical Structure for Parameter Determination.....	46
3.9	COMSOL Simulation and Numerical Analysis.....	55
3.9.1	Introduction.....	55
3.9.2	Simulation Geometry	55
3.9.3	Material Properties.....	58
3.9.4	Boundary Conditions and Physics Setup	59
3.9.5	Simulation Cases.....	59
3.9.6	Magnetic Flux Distribution.....	62
3.9.7	Simulation Limitations.....	67
3.10	Fabricated Prototype for the Mechanical Model	68
4	Analysis and Results	70
4.1	Simulation Results	70
4.1.1	Flux Linkage Variation	70
4.1.2	Induced Voltage Simulation	73
4.1.3	Key Simulation Findings	75
5	Conclusion and Futureworks	77
5.1	Conclusion	77

5.2	Future Work.....	78
6	Reference list	80
	Appendices.....	81
	Appendix 1: Arduino Code for vehicle data acquisition	81
	Appendix 2: MATLAB codes of the Analytical model.....	83
	Appendix 3: Exploded View of the EMEH Prototype.....	1
	Appendix 4: Production Drawing of base with guide shaft component	2
	Appendix 5: Production Drawing of Compression Spring component	3
	Appendix 6: Production Drawing of Bottom spring seat component.....	4
	Appendix 7: Production Drawing of Top spring seat component	5
	Appendix 8: Production Drawing of Bottom casing component.....	6
	Appendix 9: Production Drawing of Top casing component	7

List of Figures

Figure 2.1: (a) Direct piezoelectric effect, (b) converse piezoelectric effect [4]	4
Figure 2.2: Cantilever beam transducer [4], [5], [6]	6
Figure 2.3 : (a) Bimorph configuration in series connection. (b) Bimorph configuration in parallel connection. (c) Unimorph configuration. [4], [5], [6].....	6
Figure 2.4: Stack piezoelectric transducer [4], [5], [6]	6
Figure 2.5: Circular diaphragm transducer (a) Front view (b) Side view [4], [5], [6].....	6
Figure 2.6: The model of a single of freedom electromagnetic vibration energy harvester [7], [8], [9]	8
Figure 2.7: Axis system and summary of accelerometer installation [8]	10
Figure 2.8: (a) RAK7200 WisNode Track Lite (b) Dragino LoRaWAN GPS Tracker LGT-92	11
Figure 3.1: Experimental setup mounted on the vehicle control arm	27
Figure 3.2: Road classification of Sri Lanka	27
Figure 3.3: (a) E road (b) A road (c) B road (d) concrete road (e) asphalt road	28
Figure 3.4: Highway (E road) road profile	34
Figure 3.5: Highway (E road) road second profile	35
Figure 3.6: A road profile	35
Figure 3.7: A road second profile	35
Figure 3.8: Concrete road profile.....	36
Figure 3.9: B road second profile	36
Figure 3.10: B road profile.....	36
Figure 3.11: Concrete road second profile.....	36
Figure 3.12: Normalized harvested power Vs k^2Qm	37
Figure 3.13: Normalized power Vs Normalized frequency.....	38
Figure 3.14: k^2 Vs Normalized frequency	39
Figure 3.15: Qm Vs Normalized frequency.....	39
Figure 3.16: ξe Vs Normalized frequency.....	40

Figure 3.17: Electromagnetic energy harvester configurations considered during concept selection: (a) linear magnet–coil spring-based harvester, (b) rotational electromagnetic harvester configuration (c) spring-supported multi-part electromagnetic harvester assembly	42
Figure 3.18: Selected concept design.....	42
Figure 3.19: a) Initial Hand sketch	44
Figure 3.20 - Section view of the electromagnetic vibration energy harvester showing the internal component arrangement.....	45
Figure 3.21 - Exploded view of the electromagnetic vibration energy harvester assembly	46
Figure 3.22: 2D axisymmetric geometry of the electromagnetic vibration energy harvester .	56
Figure 3.23: Mesh distribution used in the COMSOL simulation.....	56
Figure 3.24: Updated 2D axisymmetric geometry of the electromagnetic vibration energy harvester used in COMSOL simulation.....	57
Figure 3.25: Parametric sweep settings used for magnet displacement in COMSOL.....	62
Figure 3.26: Magnetic flux density distribution with casing	63
Figure 3.27: Magnetic flux density distribution without casing	63
Figure 3.28: Flux linkage comparison between casing and no-casing configurations	64
Figure 3.29: Magnetic flux density distribution with ferromagnetic core	65
Figure 3.30: Magnetic flux density distribution of the updated electromagnetic harvester configuration	66
Figure 3.31: 3D magnetic flux density distribution of the updated electromagnetic vibration energy harvester model	66
Figure 3.32: Section view of the 3D model	68
Figure 3.33: Fabricated mechanical model.....	68
Figure 4.1: Flux linkage variation with magnet displacement for core and core-less configurations	71
Figure 4.2: Time-dependent simulation settings used for sinusoidal vibration input.....	73
Figure 4.3: Induced voltage waveform obtained from the time-dependent COMSOL simulation.....	74

List of Tables

Table 2.1: Piezoelectric materials and their properties [4], [5].....	5
Table 2.2: Suitability for MEMS Application of Transduction Mechanisms [7], [8]	9
Table 3.1: Tracker duty-cycle values.....	14
Table 3.2: Equation variables and their units.....	16
Table 3.3: Equation variables in electromagnetic model.....	20
Table 3.4: Road conditions where the experiment was done and the corresponding vehicle speed ranges	28
Table 3.5: Input parameters used for the analytical model.....	37
Table 3.6: Main design requirements.....	41
Table 3.7 - Key design considerations for selecting the linear electromagnetic energy harvester concept	43
Table 3.8: Components of the harvester and their functions	44
Table 3.9: Equation parameters	46
Table 3.10: Coil design parameters	52
Table 3.11: Materials used in simulations	58
Table 3.12: COMSOL simulation cases	60
Table 3.13: Summary of key COMSOL simulation parameters.....	60
Table 4.1: Sample flux linkage comparison between core and core-less configurations	70
Table 4.2: Selected coil concatenated flux values at different magnet z-positions for the updated harvester configuration.....	72
Table 4.3: Sample induced voltage values obtained from the time-dependent simulation of the updated harvester configuration.....	74
Table 4.4: Simulation main findings.....	75

List of abbreviations

AlN	Aluminum Nitride
CAD	Computer-Aided Design
DFT	Discrete Fourier Transform
EVEH	Electromagnetic Vibration Energy Harvester
FFT	Fast Fourier Transform
GPS	Global Positioning System
MEMS	Micro-Electro-Mechanical Systems
NdFeB	Neodymium Iron Boron
PLA	Polylactic Acid
PSD	Power Spectral Density
PVDF	Polyvinylidene Fluoride
PZT	Lead Zirconate Titanate
SSHI	Synchronized Switch Harvesting on Inductor

List of Appendices

Appendix 1	Arduino Code for vehicle data acquisition	Pg. 81
Appendix 2	MATLAB codes of the Analytical model	Pg. 83
Appendix 3	Exploded View of the EMEH Prototype	Pg. 84
Appendix 4	Production Drawing of base with guide shaft component	Pg. 85
Appendix 5	Production Drawing of Compression Spring component	Pg. 86
Appendix 6	Production Drawing of Bottom spring seat component	Pg. 87
Appendix 7	Production Drawing of Top spring seat component	Pg. 88
Appendix 8	Production Drawing of Bottom casing component	Pg. 89
Appendix 9	Production Drawing of Top casing component	Pg. 90

1 Introduction

1.1 Background

The use of low-power electronic tracking devices is increasing day by day. Because of this, there is a need for power sources that are small, reliable and have a long life. Most tracking devices use batteries, but a battery has only a limited amount of energy. So after some time, the battery must be charged again or replaced with a new one. This is a big problem when the device is fixed in a place that is hard to reach, or when the device must work for a long time without stopping. For this reason, energy harvesting is a good engineering solution to make the working life of low-power devices longer.[1]

There are many types of energy in the environment, but mechanical vibration is very useful for devices that are placed on vehicles. This is because a vehicle vibrates all the time when it is moving. The vibration comes from the rough road, the suspension, the contact between the tyre and the road, the engine and the movement of the chassis. Most of this vibration energy is wasted as heat in the suspension parts. If we can change a small part of this wasted energy into electrical energy, we can use it to power or charge a small electronic device.[2]

This project is about the design and development of an electromagnetic vibration energy harvesting system for a global positioning system (GPS) tracking unit that is fixed on a vehicle. The system is planned to be mounted on the lower control arm of the vehicle, because there is a lot of up and down vibration in that area when the vehicle is on the road. The electrical energy from the harvester will be used to charge the battery of the GPS tracker, so the tracker can work for a longer time.[3]

1.2 Problem Statement

A GPS tracking unit needs electrical energy to find the position from satellites, to process the data, to send the data, and to stay in standby mode. Even if we reduce the power use by duty cycling, the battery still needs to be charged after some time. In some cases, the tracker is hidden inside the vehicle or fixed in a place that is hard to reach. In such cases, charging the battery by hand is not easy.

The lower control arm of the vehicle is vibrating all the time when the vehicle is moving. But this vibration energy is not used for anything. So, the main engineering problem is how to make

a small harvester that can change this vibration into useful electrical energy. The harvester must work near the main vibration frequencies of the lower arm, and it must give enough voltage to charge a battery through a charging circuit.

Therefore, this project deals with the design, modelling, simulation, fabrication and testing of an electromagnetic vibration energy harvester (EVEH). The harvester must work under real vehicle vibration and must be able to give enough energy for a GPS tracking unit.

1.3 Aim

The aim of this project is to design, develop, and evaluate an electromagnetic vibration energy harvesting system capable of converting vehicle lower-arm vibration into usable electrical energy for recharging the battery of a low-power GPS tracking unit.

1.4 Objectives

- To conduct a comparative analysis of vehicle vibration characteristics and different energy harvesting transduction mechanisms and select a suitable transduction mechanism.
- To design a suitable vibration energy harvester based on a selected transduction mechanism.
- To fabricate a functional prototype of the designed system and conduct experimental tests to characterize its power output, efficiency, and frequency response.
- To evaluate the overall feasibility of the developed system to reliably power the tracking device and to provide recommendations for further optimization.

1.5 Scope of the Project

This project includes the design and development of a prototype EVEH. The project covers the literature review, vibration data collection, analytical modelling, computer simulation, CAD model, material selection, prototype fabrication and experimental testing.

The harvester is designed to be fixed on the lower arm of a vehicle, and it uses the up and down vibration as the input. The electrical output from the harvester is changed into DC using a rectifier and is then stored in a rechargeable battery. This battery is used to power the GPS tracking unit. The main focus of the project is the mechanical and electromechanical design of

the harvester. The battery charging circuit and the energy management system are only looked at as part of the full system.

This project does not include mass production, the design of a commercial outer case, or full military level environmental testing. It is only focused on testing at the prototype level, and comparing the results from the analytical model, the simulation and the experiments.

1.6 Current Project Progress

At this stage, most of the design and modelling work is already finished. The literature review is complete, and the electromagnetic method has been chosen as the most suitable method for this application. Real vibration data was collected from the lower control arm of a Toyota Raize by using an accelerometer with a data acquisition system. After that, the acceleration data was analysed in MATLAB to find the main vibration frequencies.

The analytical model of the harvester is also fully done. The model treats the harvester as a mass spring damper system together with an electromagnetic induction system. Calculations and parametric studies were done to see how the output of the system changes with different parameters such as mass, stiffness, damping, coil resistance, load resistance, excitation frequency and magnetic coupling.

The full design of the harvester is now finished. This includes the CAD model, the selection of materials, the selection of components and the calculation of the design parameters. COMSOL Multiphysics simulations were also done to study the change in magnetic flux, the induced voltage and the overall electromagnetic behavior. The design was changed many times to get a higher output voltage and to make the natural frequency close to the main vibration frequency.

The fabrication of the prototype has started, and some parts of the device are already made. The remaining work is to develop a custom GPS tracker, to assemble the full prototype, to test the output voltage under controlled vibration, to tune the system to give the maximum output, and finally to compare the experimental results with the analytical and simulation results.

2 Literature Review

2.1 Piezoelectric Energy Harvesting Mechanism

Piezoelectric energy harvesting was reviewed as one possible transduction mechanism for converting vehicle vibrations into usable electrical energy. Piezoelectric materials generate an electric charge when mechanical stress or deformation is applied. This is known as the direct piezoelectric effect, while the reverse process, where an electric field causes mechanical deformation, is known as the converse piezoelectric effect. These effects make piezoelectric materials suitable for harvesting vibration energy from mechanical systems such as vehicles, machinery, and structures.

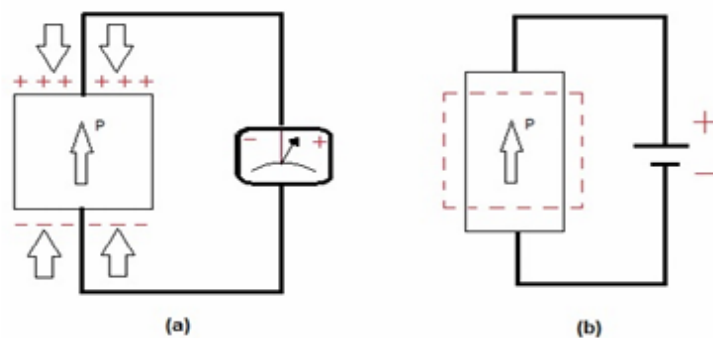


Figure 2.1: (a) Direct piezoelectric effect, (b) converse piezoelectric effect [4]

The literature review considered the main operating principles, material types, structural configurations, operating modes, power output ranges, and energy extraction methods related to piezoelectric harvesters. Common piezoelectric materials such as Lead Zirconate Titanate (PZT), Aluminum Nitride (AlN) and Polyvinylidene fluoride (PVDF) were reviewed. Among these, PZT was identified as a strong material option because of its higher piezoelectric coefficients and better energy conversion capability compared with polymer-based materials such as PVDF. However, material flexibility, durability, cost, and operating environment must also be considered when selecting a suitable material for a vibration energy harvesting system.

Table 2.1: Piezoelectric materials and their properties [4], [5]

Material Category	Example Materials	Properties
Piezo-Ceramics	Lead Zirconate Titanate (PZT) Barium Titanate (BaTiO ₃) Zinc Oxide (ZnO)	PZT is the most common and active human-made material. PZT film coefficients are at least two times greater than PVDF and BaTiO ₃ , generating higher power output. PZT can generate sufficient piezoelectricity when deformed by only 0.1%. PZT has higher power output per unit area under high temperatures compared to PVDF. For energy harvesting, PZT-5H is considered a strong choice due to its high d_{33} value, availability, and low cost.
Piezo-Polymers	Polyvinylidene Fluoride (PVDF)	Polymers generally yield lower power output. For instance, PZT yielded a harvested power output per unit area of $9.37 \times 10^{-3} \text{ mW/mm}^2$ in one study, compared to $5.31 \times 10^{-4} \text{ mW/mm}^2$ for PVDF.

Different piezoelectric transducer configurations were also reviewed, including cantilever, stacked, cymbal, and circular diaphragm structures. Cantilever-based harvesters are commonly used for low-frequency ambient vibrations because of their simple design and high strain generation. Stacked piezoelectric configurations are more suitable for high-force applications, such as vehicle suspension and machinery vibration, because they can withstand larger mechanical loads and generate higher power output. Cymbal and circular diaphragm structures are also useful in specific applications where space limitation, amplified strain, or pressure-based excitation is important.

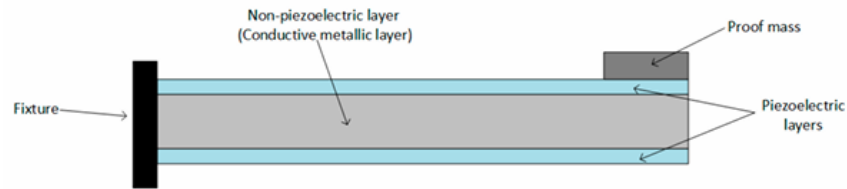


Figure 2.2: Cantilever beam transducer [4], [5], [6]

15

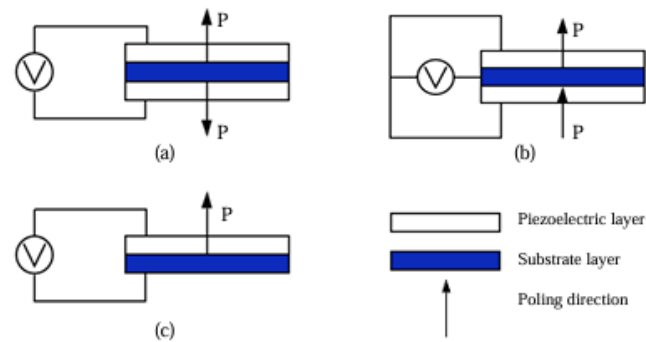


Figure 2.3 : (a) Bimorph configuration in series connection. (b) Bimorph configuration in parallel connection. (c) Unimorph configuration. [4], [5], [6]

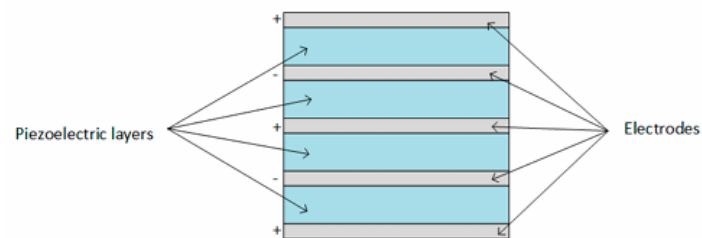


Figure 2.4: Stack piezoelectric transducer [4], [5], [6]

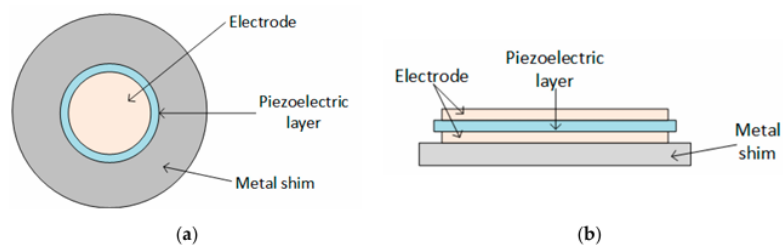


Figure 2.5: Circular diaphragm transducer
(a) Front view (b) Side view [4], [5], [6]

The review also showed that the performance of piezoelectric harvesters strongly depends on vibration frequency, resonance matching, mechanical loading, and electrical energy extraction

circuits. Piezoelectric devices normally produce maximum output when their natural frequency matches the input vibration frequency. However, vehicle vibrations are not constant and can vary over a wide frequency range. Therefore, resonance tuning and bandwidth improvement methods are important for improving the practical performance of piezoelectric harvesters under real vehicle vibration conditions. The reviewed literature also discussed methods such as Synchronized Switch Harvesting on Inductor (SSHI) circuits to improve energy extraction efficiency under variable vibration conditions. [6], [7]

Overall, the piezoelectric literature review showed that piezoelectric energy harvesting is compact, simple, and suitable for low-power applications such as sensors and tracking devices. However, its performance is highly dependent on material selection, structural configuration, resonance matching, and energy management circuits. The electromagnetic transduction mechanism was also reviewed separately by another group member. After comparing the piezoelectric and electromagnetic mechanisms, the electromagnetic mechanism was selected as the final transduction method for the vibration energy harvesting system because it was more suitable for the expected vehicle vibration conditions and project requirements.

2.2 Electromagnetic Energy Harvesting Mechanism

Electromagnetic energy harvesting was reviewed as another possible method for converting vehicle vibration into electrical energy. This method works based on Faraday's law, where voltage is induced when there is relative motion between a permanent magnet and a coil. In a vibration energy harvester, the moving mass vibrates with the vehicle excitation, causing the magnet and coil to move relative to each other and generate an alternating voltage.

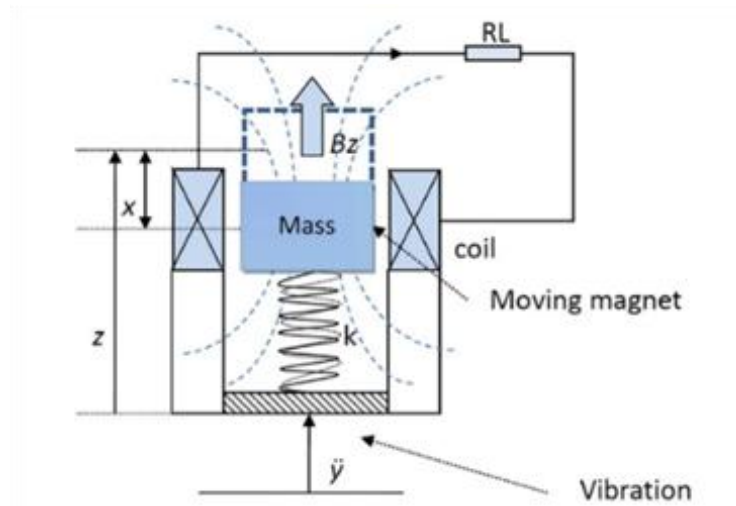


Figure 2.6: The model of a single of freedom electromagnetic vibration energy harvester [7], [8], [9]

Traditional vibration energy harvesters are usually designed to operate near resonance, where the excitation frequency matches the natural frequency of the device. This improves energy conversion, but it also creates a narrow operating bandwidth. This is a limitation for vehicle applications because vehicle vibration changes with road surface, speed, and suspension behaviour.

Recent studies therefore focus on broadband and non-resonant electromagnetic vibration energy harvesters. These systems can operate over a wider frequency range and are useful for low-frequency vibration sources such as vehicles, machinery, and infrastructure. Electromagnetic harvesters are mechanically robust and suitable for low-frequency vibration, but their output voltage can be low at small scale and depends strongly on magnet strength, coil design, relative displacement, and load matching.

For this project, electromagnetic harvesting was considered suitable because the vehicle lower control arm provides low-frequency vibration, and the prototype can be built using simple components such as magnets, copper coils, springs, guide shafts, and a polymer casing. [8], [9], [10]

2.3 Comparison of Transduction Mechanisms and Selection Justification

After reviewing piezoelectric, electromagnetic, and electrostatic energy harvesting methods, the mechanisms were compared to select the most suitable option for this project.

Table 2.2: Suitability for MEMS Application of Transduction Mechanisms [7], [8]

Transduction Method	Suitability for MEMS Applications
Electromagnetic	Efficient for larger-scale systems, but difficult to scale down and integrate efficiently into silicon-based MEMS technology. However, it is suitable for this prototype because it is robust and works well with low-frequency vibration.
Piezoelectric	Suitable for low-power applications and can generate tens to hundreds of microwatts. However, cantilever structures may face reliability issues due to fatigue or breakage under repeated vibration.

Although MEMS suitability was considered, this project mainly focuses on a practical vehicle-mounted prototype rather than a silicon-based MEMS device. Piezoelectric harvesting is compact, but it depends strongly on strain, resonance matching, and material durability.

Therefore, electromagnetic vibration energy harvesting was selected as the final transduction mechanism. It is more suitable for low-frequency vehicle vibration, does not require an external bias voltage, and can be fabricated using simple and robust components. The selected final concept is a linear electromagnetic vibration energy harvester, which will be used for analytical modelling, COMSOL simulation, prototype fabrication, and testing. [8], [9]

2.4 Vehicle Vibration Measurement and Data Analysis for Energy Harvesting

Vehicle vibration measurement is important because the harvester should be designed using the actual vibration available at the selected mounting location. A study conducted at the University of Michigan Transportation Research Institute used an instrumented Infiniti QX56 SUV to measure how road-surface excitations are transmitted through the vehicle structure. Although the study mainly focused on rumble-strip detection, its accelerometer arrangement and signal processing method are useful for vibration energy harvesting applications.

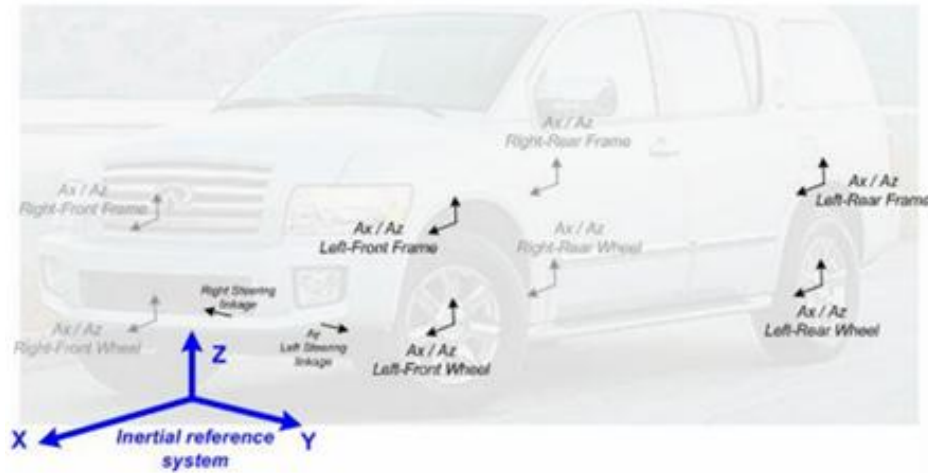


Figure 2.7: Axis system and summary of accelerometer installation [8]

The study used accelerometers on wheel stations, suspension arms, chassis frame, and steering tie-rod ends to measure longitudinal, vertical, and lateral acceleration responses. The results showed that vibration response depends strongly on sensor location, road surface, vehicle speed, and operating condition. Milled-in rumble strips produced stronger periodic vibration, while smooth concrete and dirt roads mainly produced broadband vibration.

For this project, these findings justify measuring real lower control arm vibration instead of assuming a fixed excitation frequency. Since the proposed electromagnetic harvester mainly uses vertical motion, the vertical acceleration component should be analysed using time-domain plots, FFT, and PSD methods to identify the dominant frequency range for harvester tuning.

2.5 GPS Tracker Selection and Energy Demand

The vibration energy harvester developed in this project is intended to supplement the internal battery of a stealth GPS tracking unit mounted on a military vehicle. A GPS tracker is a portable device that determines its position by receiving radio-frequency signals from at least four orbiting satellites and applying trilateration to compute latitude, longitude, elevation, and time [11]. A typical unit comprises a GPS receiver, a communication module (GSM, LoRaWAN, or satellite), a rechargeable battery, GPS and communication antennas, a microcontroller, and onboard memory for offline logging [12]. Because the harvester output is small and intermittent, the choice of tracker has a direct influence on whether harvested energy can meaningfully extend the autonomous operating life of the system. This section reviews the

available tracker categories, identifies a suitable device for the application, and establishes the energy demand benchmark used in later chapters.

2.5.1 Types of GPS Trackers and Suitability for VEH Applications

The literature distinguishes five common tracker categories: Bluetooth, OBD, satellite, wired, and wireless [13]. Bluetooth trackers are limited to very short range and are unsuitable for vehicle tracking. OBD and wired trackers draw continuous power from the vehicle electrical system, which contradicts the stealth requirement and removes the rationale for energy harvesting. Satellite trackers offer global coverage but depend on expensive hardware and subscription services. Battery-powered wireless trackers offer the most compatible match for a vibration energy harvesting application because they are compact, can be installed discreetly, and rely on internal rechargeable batteries that can be supplemented by harvested energy.

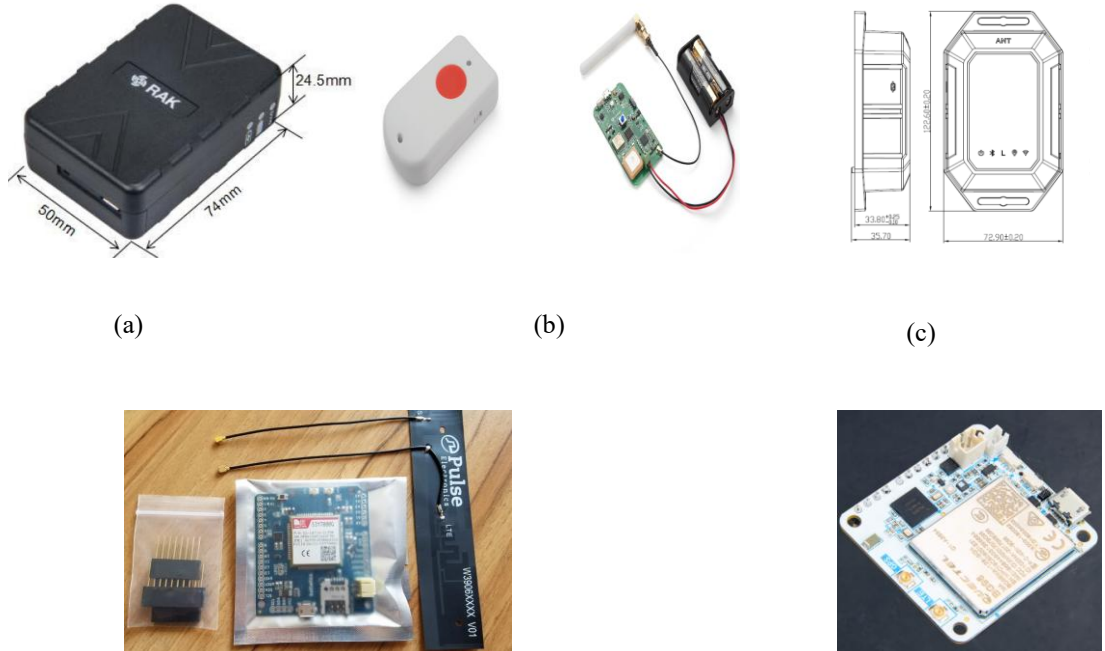


Figure 2.8: (a) RAK7200 WisNode Track Lite (b) Dragino LoRaWAN GPS Tracker LGT-92

(c) LW001-BG PRO (d) Botletics™ SIM7000 LTE CAT-M1 (e) RAK 5010

2.5.2 Selection of a Suitable Wireless Tracker

Three representative wireless trackers, the Dragino LGT-92, the RAK7200, and the SIM7000 NB-IoT module, were compared in the literature against the parameters most relevant to a low-

power harvester source: sleep current, rechargeable battery support, operating voltage range, charging interface, GPS-fix current, transmission peak current, and enclosure protection [14], [15]. The Dragino LGT-92 was identified as the most suitable device, offering an ultra-low sleep current in backup mode, a 1000 mAh Li-ion rechargeable battery, micro-USB charging compatible with a rectified harvester output, LoRaWAN communication, and a compact form factor under 55 g [16]. These features make it the most practical platform for evaluating supplementary charging from a vibration energy harvester.

2.6 Research Gap

Based on the literature review, the following research gaps were identified:

1. Many vibration energy harvesting studies focus on ideal laboratory excitation, but real vehicle lower-control-arm vibration is broadband, irregular, and road-dependent.
2. Piezoelectric harvesters are compact and widely studied, but their practical use in vehicle suspension environments can be limited by force transfer, material brittleness, and resonance mismatch.
3. Electromagnetic harvesters are suitable for low-frequency vibration, but their output voltage can be low unless magnet strength, coil design, relative displacement, and load matching are optimized.
4. Many studies evaluate harvester output alone, but fewer studies connect the harvested energy to the actual duty-cycle energy requirement of a GPS tracking unit.
5. The energy management stage is often simplified, although rectification losses, converter efficiency, and battery charging limits are critical for low-power harvesters.
6. A practical prototype must consider not only theoretical power output but also mass, spring stiffness, alignment, friction, casing material, coil resistance, and manufacturability.

These gaps show the need for a complete system-level study that includes vibration measurement, electromagnetic harvester design, analytical modelling, simulation, prototype fabrication, GPS tracker energy analysis, and experimental validation.

3 Methodology

3.1 Introduction

This chapter presents the methodology followed to design, model, simulate, fabricate and validate the electromagnetic vibration energy harvesting system developed for a vehicle-mounted GPS tracking unit. The methodology was structured such that the final design was not based only on theoretical assumptions.

First, the vibration source and power requirement of the GPS tracking unit were identified. Then, different vibration energy harvesting mechanisms were reviewed and compared. After selecting the electromagnetic transduction mechanism, real vibration data were collected from the lower control arm of a Toyota Raize using an accelerometer-based data acquisition system. The measured data were processed in MATLAB to identify the dominant excitation frequencies and acceleration amplitudes. These values were then used to develop the analytical model, select design parameters, conduct COMSOL simulation, fabricate the prototype and prepare the experimental testing procedure.

3.2 Calculation power requirement of the GPS tracking unit

The GPS tracker is operated in duty-cycle mode to reduce average power consumption. In this method, the tracker stays in sleep mode for most of the time and wakes up only to get GPS location and transmit data.

For this project, one duty cycle is,

$$T = 600 \text{ s} \quad (3.1)$$

This means the tracker sends location data once every 10 minutes.

The total energy per cycle is,

$$E_{Total} = E_{Sleep} + E_{Active} \quad (3.2)$$

The average current is calculated using,

$$I_{avg} = \frac{I_{sleep}t_{sleep} + I_{fix}t_{fix} + I_{tx,total}t_{tx}}{600} \quad (3.3)$$

The values used in the project are shown below.

Table 3.1: Tracker duty-cycle values

Parameter	Value
Cycle time	600 s
Sleep current	0.085 mA
Sleep time	550.9032 s
GPS fix current	54 mA
GPS fix time	35 s
Total transmission current	133 mA
Transmission time	14.0968 s
Average current	6.35 mA

The charge consumed per 10-minute cycle is,

$$Q_{cycle} = I_{avg} \frac{600}{3600} \quad (3.4)$$

$$Q_{cycle} = 6.35 \times \frac{1}{6} \quad (3.5)$$

$$Q_{cycle} = 1.0583 \text{ mAh} \quad (3.6)$$

For a 1000 mAh battery, the estimated battery life without harvesting is,

$$L = \frac{1000}{6.35} \quad (3.7)$$

$$L = 157.48 \text{ h} \quad (3.8)$$

$$L = 6.56 \text{ days} \quad (3.9)$$

The energy consumed per cycle is,

$$E_{cycle} = VIt \quad (3.10)$$

$$E_{cycle} = 3.7 \times \frac{6.35}{1000} \times 600 \quad (3.11)$$

$$E_{cycle} = 14.097 J \quad (3.12)$$

These results show that duty-cycle operation greatly reduces the average power demand of the GPS tracker. Therefore, the harvester does not need to power the tracker continuously. It can provide supplementary energy to extend battery life.

3.3 Selection of Transduction Mechanism

The selection of the transduction mechanism is important because it directly affects the harvester output, size, durability, fabrication complexity and suitability for vehicle vibration conditions. Three main energy harvesting methods considered were piezoelectric, electromagnetic and electrostatic harvesting.

Piezoelectric harvesting was considered because it is compact and suitable for low-power sensor applications. However, its output depends strongly on mechanical strain, loading condition and resonance matching. Since the lower control arm vibration is irregular and low-frequency, piezoelectric stacks may require high force while cantilever-type designs may face durability and tuning issues.

Electrostatic harvesting was also considered, but it usually requires an initial bias voltage or charged capacitor system. This increases circuit complexity and makes it less suitable for a compact, self-contained tracking unit.

While efficient for macroscopic systems, electromagnetic harvesting mechanism can be fabricated using simple and robust components. Therefore, electromagnetic energy harvesting was selected as the final method.

3.4 Analytical Model Development

3.4.1 Introduction

After selecting the transduction mechanism, the analytical model was developed for the electromagnetic vibration energy harvesting system. The purpose of the model is to describe

the relationship between the vehicle lower control arm vibration input, the mechanical response of the harvester, the electromagnetic induction process and the electrical power delivered to an external load.

The harvester is modelled as a base-excited single-degree-of-freedom mass-spring-damper system coupled with an electromagnetic transducer. This modelling approach is commonly used for inertial vibration energy harvesters where a dynamic mass moves relative to the harvester housing and the relative motion between a magnet and a coil generates voltage. The model is used to determine the required moving mass, spring stiffness, damping coefficient, electromagnetic coefficient, load resistance, coil inductance, coil resistance and expected harvested power.

The analytical model was also used as the basis for the MATLAB calculations and parameter calculations carried out during the design stage. The results from this chapter provide the theoretical foundation for the design parameter selection.

3.4.2 Mechanical Model

The electromagnetic vibration energy harvester is represented as an inertial generator mounted on a vibrating base. The base represents the vehicle lower control arm while the moving mass represents the magnet assembly or moving mass of the harvester. The relative motion between the mass and the base produces a time varying magnetic flux linkage through the coil.

Table 3.2: Equation variables and their units

Symbol	Description	Unit
M	Seismic mass / dynamic mass	kg
K	Spring stiffness	N/m
D	Mechanical damping coefficient	Ns/m
$y(t)$	Base displacement input	m
$u(t)$	Relative displacement between mass and base	m

$x(t)$	Absolute displacement of seismic mass	m
--------	---------------------------------------	---

The absolute displacement of the dynamic mass is given by,

$$x(t) = y(t) + u(t) \quad (3.13)$$

For base excitation, the equation of motion of the mass-spring-damper system can be written as,

$$M\ddot{u}(t) + D\dot{u}(t) + Ku(t) = -M\ddot{y}(t) \quad (3.14)$$

Equation (3.14) shows that the excitation input to the harvester is the base acceleration $\ddot{y}(t)$ rather than a directly applied external force.

When electromagnetic coupling is included, the current generated in the coil produces an opposing electromagnetic force. Therefore, the coupled mechanical equation becomes,

$$M\ddot{u}(t) + D\dot{u}(t) + Ku(t) + \beta i(t) = -M\ddot{y}(t) \quad (3.15)$$

where β is the electromechanical coupling coefficient and $i(t)$ is the induced current in the coil.

3.4.3 Natural Frequency and Spring Stiffness Selection

The harvester is designed to operate near resonance because the relative displacement between the magnet and coil is maximized close to the natural frequency of the mechanical system. Since the harvested power is proportional to the square of the relative displacement amplitude, resonance operation significantly improves the electrical power output. The undamped natural angular frequency is,

$$\omega_0 = \sqrt{\frac{K}{M}} \quad (3.16)$$

The corresponding natural frequency in hertz is,

$$f_0 = \frac{\omega_0}{2\pi} \quad (3.17)$$

Rearranging (3.16), the required spring stiffness for a selected target frequency is,

$$K = M\omega_0^2 = M(2\pi f_0)^2 \quad (3.18)$$

This stiffness value is used as the theoretical target for selecting and designing the spring arrangement of the prototype.

3.4.4 Damping ratio and Mechanical Quality Factor

The damping ratio of the harvester determines the sharpness of the resonance peak and the useful frequency bandwidth of the device. For a single-degree-of-freedom system, the damping ratio is defined as,

$$\zeta = \frac{D}{2\sqrt{KM}} \quad (3.19)$$

The mechanical quality factor is,

$$Q_m = \frac{1}{2\zeta} \quad (3.20)$$

So,

$$Q_m = \frac{\sqrt{KM}}{D} = \frac{M\omega_0}{D} \quad (3.21)$$

The damping coefficient can be calculated from,

$$D = \frac{M\omega_0}{Q_m} \quad (3.22)$$

A lightly damped system is suitable for resonance-based energy harvesting. However, a very low damping ratio also produces a narrow operating bandwidth. Therefore, the system must be tuned close to the dominant vibration frequency obtained from the vehicle vibration data.

3.4.5 Frequency Response of the Mechanical System

For harmonic base excitation, the base displacement is represented as,

$$y(t) = Y\sin(\omega t) \quad (3.23)$$

where Y is the base displacement amplitude and ω is the excitation angular frequency.

The base acceleration amplitude is,

$$A = \omega^2 Y \quad (3.24)$$

The frequency ratio is defined as,

$$r = \frac{\omega}{\omega_0} \quad (3.25)$$

For the uncoupled mechanical system, the steady-state relative displacement amplitude is,

$$U = \frac{A/\omega_0^2}{\sqrt{(1-r^2)^2 + \left(\frac{r}{Q_m}\right)^2}} \quad (3.26)$$

At resonance, $r = 1$, the approximate relative displacement amplitude becomes,

$$U_{\text{res}} \approx \frac{AQ_m}{\omega_0^2} \quad (3.27)$$

Equation (3.27) shows that the relative displacement increases with excitation acceleration and mechanical quality factor but decreases with the square of natural angular frequency. This relationship is important because electromagnetic induction depends directly on the relative velocity between the magnet and coil.

The relative velocity amplitude is,

$$V_u = \omega U \quad (3.28)$$

Since the induced voltage is proportional to relative velocity, the harvester output increases when the system produces larger relative motion at or near resonance.

3.4.6 Electromagnetic Model

The electromagnetic transducer consists of a permanent magnet and a coil. When relative motion occurs between the magnet and coil, the magnetic flux linkage through the coil changes with time. According to Faraday's law of electromagnetic induction, the induced voltage is proportional to the rate of change of flux linkage.

The induced open-circuit voltage can be written as,

$$v_{oc}(t) = \frac{d\lambda}{dt} \quad (3.29)$$

where λ is the flux linkage.

Since the flux linkage varies with relative displacement u ,

$$\frac{d\lambda}{dt} = \frac{d\lambda}{du} \frac{du}{dt} \quad (3.30)$$

The electromechanical coupling coefficient is therefore defined as,

$$\beta = \frac{d\lambda}{du} \quad (3.31)$$

So, the induced open circuit voltage becomes,

$$v_{oc}(t) = \beta \dot{u}(t) \quad (3.32)$$

The electrical circuit of the harvester is represented by a voltage source in series with the coil resistance, coil inductance and external load resistance.

Table 3.3: Equation variables in electromagnetic model

Symbol	Description	Unit
L_0	Coil inductance	H
r_L	Coil internal resistance	Ω
R	External load resistance	Ω
$i(t)$	Coil current	A
$v(t)$	Load voltage	V

The electrical governing equation is,

$$L_0 \frac{di(t)}{dt} + (r_L + R)i(t) = \beta \dot{u}(t) \quad (3.33)$$

Therefore, the final coupled mechanical-electrical model is,

$$M\ddot{u}(t) + D\dot{u}(t) + Ku(t) + \beta i(t) = -M\ddot{y}(t) \quad (3.34)$$

$$L_0 \frac{di(t)}{dt} + (r_L + R)i(t) = \beta \dot{u}(t) \quad (3.35)$$

Equations (3.34) and (3.35) describe the complete analytical behavior of the electromagnetic vibration energy harvester.

3.4.7 Dimensionless Parameters

To generalize the model and compare the effect of different design parameters, the equations are normalized using the natural angular frequency,

$$\omega_0 = \sqrt{\frac{K}{M}} \quad (3.36)$$

The normalized time is,

$$\tau = \omega_0 t \quad (3.37)$$

The main dimensionless parameters are defined as follows,

$$Q_m = \frac{\sqrt{KM}}{D} \quad (3.38)$$

$$\xi_e = \frac{r_L}{2\omega_0 L_0} \quad (3.39)$$

$$\xi_c = \frac{R}{2\omega_0 L_0} \quad (3.40)$$

$$k^2 = \frac{\beta^2}{KL_0} \quad (3.41)$$

where Q_m is the mechanical quality factor, ξ_e is the resistive loss coefficient, ξ_c is the load coefficient and k^2 is the squared electromechanical coupling coefficient. These normalized parameters allow the harvester behaviour to be studied without depending only on one particular coil or magnet geometry.

3.4.8 Normalized Governing Equations at Non-resonant Excitation

To analyse the harvester under general vibration conditions, the coupled mechanical and electrical equations are normalized using the natural angular frequency of the mechanical oscillator. This is important because the vehicle lower control arm vibration is not perfectly sinusoidal at one fixed resonant frequency. Instead, the measured excitation contains different frequency components depending on road surface and vehicle motion. Therefore, the analytical model must also describe the harvester response when the excitation frequency is not equal to the natural frequency.

The natural angular frequency is defined as,

$$\omega_0 = \sqrt{\frac{K}{M}} \quad (3.42)$$

The normalized excitation frequency is,

$$\Omega = \frac{\omega}{\omega_0} \quad (3.43)$$

where ω is the angular frequency of the external vibration and ω_0 is the natural angular frequency of the harvester.

Using the dimensionless parameter definitions, the normalized mechanical equation becomes,

$$y'' = u'' + \frac{u'}{Q_m} + u + k^2 i \quad (3.44)$$

The normalized electrical equation becomes,

$$v = u' - 2\xi_e i - i' \quad (3.45)$$

where the prime symbol (') denotes differentiation with respect to normalized time τ .

For a resistive load, the load voltage is related to the current by,

$$v = 2\xi_c i \quad (3.46)$$

Substituting this into the normalized electrical equation gives,

$$2\xi_c i = u' - 2\xi_e i - i' \quad (3.47)$$

$$i' + 2(\xi_e + \xi_c)i = u' \quad (3.48)$$

For harmonic excitation at a normalized angular frequency Ω , the displacement and current can be represented in the Fourier domain as $\hat{u}$ and $\hat{i}$. Therefore,

$$i' \rightarrow j\Omega \hat{i} \quad (3.49)$$

$$u' \rightarrow j\Omega \hat{u} \quad (3.50)$$

So the current in the Fourier domain is,

$$j\Omega \hat{i} + 2(\xi_e + \xi_c)\hat{i} = j\Omega \hat{u} \quad (3.51)$$

$$\hat{i} = \frac{j\Omega}{j\Omega + 2(\xi_e + \xi_c)} \hat{u} \quad (3.52)$$

Let,

$$\alpha = \xi_e + \xi_c \quad (3.53)$$

Then,

$$\hat{i} = \frac{j\Omega}{j\Omega + 2\alpha} \hat{u} \quad (3.54)$$

The magnitude of the normalized current is,

$$i_M = \frac{\Omega}{\sqrt{\Omega^2 + 4\alpha^2}} u_M \quad (3.55)$$

The mechanical equation in the Fourier domain is written as,

$$-\Omega^2 \hat{y} = -\Omega^2 \hat{u} + \frac{j\Omega}{Q_m} \hat{u} + \hat{u} + k^2 \hat{i} \quad (3.56)$$

Substituting the expression for $\hat{t}$,

$$-\Omega^2 \hat{y} = \left[1 - \Omega^2 + \frac{j\Omega}{Q_m} + k^2 \left(\frac{j\Omega}{j\Omega + 2\alpha} \right) \right] \hat{u} \quad (3.57)$$

The term involving electromagnetic coupling can be separated into real and imaginary parts,

$$\frac{j\Omega}{j\Omega + 2\alpha} = \frac{\Omega^2 + j2\alpha\Omega}{\Omega^2 + 4\alpha^2} \quad (3.58)$$

Therefore, the magnitude of the relative displacement is,

$$u_M = \frac{\Gamma_M / \omega_0^2}{\sqrt{\left[1 - \Omega^2 + \frac{k^2 \Omega^2}{\Omega^2 + 4\alpha^2} \right]^2 + \left[\frac{\Omega}{Q_m} + \frac{2k^2 \alpha \Omega}{\Omega^2 + 4\alpha^2} \right]^2}} \quad (3.59)$$

where Γ_M is the physical base acceleration amplitude.

The harvested power across the load resistance is,

$$P_R = \frac{1}{2} R I_M^2 \quad (3.60)$$

Using the normalized current expression, the harvested power becomes,

$$P_R = \frac{k^2 \xi_c M \omega_0^3 \Omega^2}{\Omega^2 + 4(\xi_e + \xi_c)^2} u_M^2 \quad (3.61)$$

Substituting the displacement magnitude into the power expression gives,

$$P_R = \frac{k^2 \xi_c M \Gamma_M^2}{\omega_0} \frac{\Omega^2}{\Omega^2 + 4\alpha^2} \frac{1}{\left[1 - \Omega^2 + \frac{k^2 \Omega^2}{\Omega^2 + 4\alpha^2} \right]^2 + \left[\frac{\Omega}{Q_m} + \frac{2k^2 \alpha \Omega}{\Omega^2 + 4\alpha^2} \right]^2} \quad (3.62)$$

The normalized harvested power can be expressed as,

$$P_{\text{norm}} = \frac{P_R}{M \Gamma_M^2 / \omega_0} \quad (3.63)$$

So,

$$P_{\text{norm}} = k^2 \xi_c \frac{\Omega^2}{\Omega^2 + 4\alpha^2} \frac{1}{\left[1 - \Omega^2 + \frac{k^2 \Omega^2}{\Omega^2 + 4\alpha^2}\right]^2 + \left[\frac{\Omega}{Q_m} + \frac{2k^2 \alpha \Omega}{\Omega^2 + 4\alpha^2}\right]^2} \quad (3.64)$$

Where,

$$\alpha = \xi_e + \xi_c \quad (3.65)$$

This expression describes the harvested power for any normalized excitation frequency Ω . When $\Omega = 1$, the excitation frequency is equal to the natural frequency and the model reduces to the resonance case. When $\Omega \neq 1$, the response decreases depending on the difference between the excitation frequency and the natural frequency. This is important for vehicle vibration energy harvesting because the lower control arm vibration varies with road profile, vehicle speed and suspension motion. Therefore, the non-resonant model provides a more realistic estimate of harvester performance under practical operating conditions.

3.4.9 Governing Equation for Power Output at Resonance

At resonance, the excitation frequency is equal to the natural frequency,

$$\omega = \omega_0 \quad (3.66)$$

The normalized harvested power at resonance for a classical resistive load can be written as,

$$P_R = \frac{k^2 Q_m \xi_c [1 + 4(\xi_e + \xi_c)^2]}{[1 + 4(\xi_e + \xi_c)^2 + 2k^2 Q_m (\xi_e + \xi_c)]^2} \left(\frac{M Q_m A^2}{\omega_0} \right) \quad (3.67)$$

The corresponding normalized power is,

$$P_{\text{norm}} = \frac{P_R}{M Q_m A^2 / \omega_0} \quad (3.68)$$

Therefore,

$$P_{\text{norm}} = \frac{k^2 Q_m \xi_c [1 + 4(\xi_e + \xi_c)^2]}{[1 + 4(\xi_e + \xi_c)^2 + 2k^2 Q_m (\xi_e + \xi_c)]^2} \quad (3.69)$$

Above equations were used to study the influence of load coefficient, resistive loss coefficient, electromechanical coupling and mechanical quality factor on harvested power.

3.4.10 Design Implications from Analytical Model

The analytical model provides several important design conclusions for the electromagnetic vibration energy harvester.

The device should be tuned close to the dominant vertical vibration frequency of the vehicle lower control arm. The selected target frequency is 11 Hz. The spring-mass system's equivalent stiffness and mass should be calculated accordingly.

The harvester should be designed to allow sufficient relative displacement between the magnet and coil. A larger relative displacement increases relative velocity and therefore increases induced voltage. However, mechanical end stops must be included to prevent damage under high vibration amplitudes.

The coil resistance should be kept as low as practically possible because internal resistance reduces the useful power delivered to the external load. The wire diameter, number of turns and coil size must therefore be selected by balancing voltage generation and resistive loss.

The analytical model shows that resonance matching, magnetic coupling, mechanical damping and electrical loading are strongly interconnected. Therefore, the final design should be validated using both COMSOL simulation and experimental testing.

3.5 Experimental Vibration Data Acquisition and Signal Processing

After selecting the electromagnetic harvesting method, real vehicle vibration data were collected from the selected mounting location. The lower control arm of a Toyota Raize was selected because it is directly connected to the suspension system and receives road induced excitation through the wheel assembly. Since the harvester operates using relative motion between the magnet and coil, the relative vibration components have to be measured.

The vibration measurement system consisted of an ADXL345 tri-axial accelerometer, a microcontroller-based data acquisition circuit, an SD card storage module, a battery supply and

connecting wires. The accelerometer was mounted securely on the lower control arm so that the measured acceleration represented the actual vibration response of the selected location.

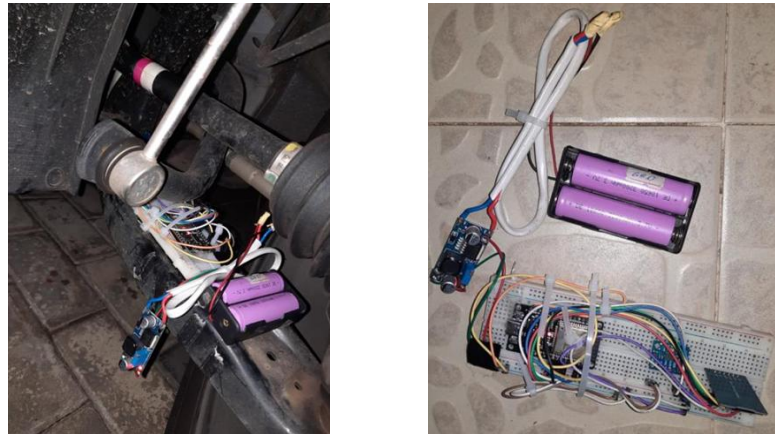


Figure 3.1: Experimental setup mounted on the vehicle control arm

The vehicle was tested under different road profiles including main road, concrete road, highway road, B-road and tar road conditions. These road profiles were selected to capture different vibration conditions and to avoid designing the harvester based on only one ideal road input. The data were recorded at different vehicle speeds, and each recording cycle was carried out for approximately 10 minutes.

Sri Lanka Road Classification			OSM Tag			Visual	
Feature Name	Implementation Entity	Description	Key	Value	Element	Rendering	Photo
Expressway(E)	Road Development Authority (RDA)	A highway especially planned for high-speed traffic, usually having few if any intersections, limited points of access or exit and A wide road (more than 30m width or less than 30m width) for high speed transportation.	highway	motorway			
Main Road(A)	Road Development Authority (RDA)	The most important roads in a country's system that aren't motorways. (Need not necessarily be a divided highway), and Inter-provincial trunk roads Connecting major cities and ports and access to link the province to province.	highway	trunk			deleted media file
Main Road(B)	Road Development Authority (RDA)	The next most important roads in a country's system. (Often link larger towns.) Intra-provincial arterial roads connecting major urban areas.	highway	primary			
Secondary/Minor Road (C & D)	Provincial Road Development Authority (PRDA) of relevant Provincial Council	The next most important roads in the country's system. (Often link smaller towns and villages.) and Minor feeder roads connecting settlements with markets and other activities places.	highway	secondary			
Jeep/Car Road	In-house Road Maintenance Unit of Municipal/Urban Council or Local Authority	Local roads providing access to specific location (Example: Roads access to religion places, historical places etc...)	highway	residential			
Jeep/Car Track		A roads providing access to plantation, forest, irrigation, etc... Usually unpaved.	highway	track			
Footpath		For designated footpaths; i.e., mainly/exclusively for pedestrians. This includes walking tracks and gravel paths.	highway	path			

Figure 3.2: Road classification of Sri Lanka

Table 3.4: Road conditions where the experiment was done and the corresponding vehicle speed ranges

Road Condition	Number of data sets	Vehicle speed (km/h)
E	2	70 - 100
A	2	40 - 60
B	2	20 - 50
C & D	2	10 - 30

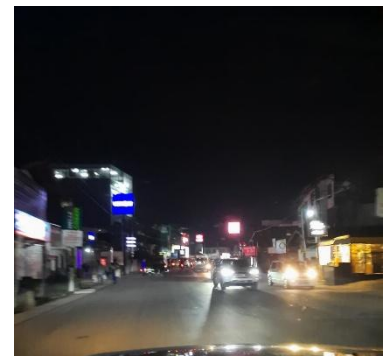
Figure 3.3 shows the road conditions used for the vibration measurement experiment. The collected acceleration data were then used for time-domain analysis, FFT analysis and PSD estimation. The vertical acceleration component was given particular attention because it directly relates to the intended operating direction of the electromagnetic vibration energy harvester.



(a)



(b)



(c)



(d)



(e)

Figure 3.3: (a) E road (b) A road (c) B road (d) concrete road (e) asphalt road

3.6 PSD Analysis using Welch's Method for Road Test Vibration Data

In addition to the Fast Fourier Transform (FFT) analysis, the measured road vibration data were analysed using Power Spectral Density (PSD). The purpose of the PSD analysis was to determine how the acceleration energy of the road induced vibration is distributed over frequency. This is important because the electromagnetic vibration energy harvester does not receive a single fixed frequency sinusoidal input during real vehicle operation. Instead, the lower control arm vibration is broadband and varies with road surface, vehicle speed, suspension response and transient impacts.

The road test data were collected using an ADXL345 accelerometer based data logging system mounted on the lower control arm of a Toyota Raize. The acceleration data were stored for MATLAB post processing and the vertical acceleration component was selected because the proposed harvester is designed to extract energy mainly from vertical vibration. This approach is also consistent with vibration energy harvester design studies where measured vehicle acceleration profiles are used to identify the frequency bands that contain useful vibration energy.

3.6.1 Gravity component removal from the data

The acceleration measured by the accelerometer contains both the road induced dynamic acceleration and the projection of gravity along the sensor axis. Therefore, the measured vertical acceleration can be expressed as,

$$a_z(t) = a_{\text{dynamic}}(t) + g_z \quad (3.70)$$

where $a_z(t)$ is the measured vertical acceleration, $a_{\text{dynamic}}(t)$ is the road induced vibration acceleration and g_z is the gravity projection on the vertical sensor axis.

The mean value of the measured acceleration was used to estimate the constant gravity and DC offset component,

$$\bar{a}_z = \frac{1}{T} \int_0^T a_z(t) dt \quad (3.71)$$

For the sampled acceleration signal, this becomes,

$$\bar{a}_z = \frac{1}{N} \sum_{n=0}^N a_z[n] \quad (3.72)$$

where N is the total number of samples. The dynamic acceleration signal used for PSD calculation was then obtained as,

$$a_{\text{dynamic}}[n] = a_z[n] - \bar{a}_z \quad (3.73)$$

For simplicity, the processed acceleration signal is denoted as,

$$a[n] = a_{\text{dynamic}}[n] \quad (3.74)$$

3.6.2 Frequency-Domain Representation

The Discrete Fourier Transform (DFT) of the processed acceleration signal is,

$$A[k] = \sum_{n=0}^{N-1} a[n] e^{-j2\pi kn/N} \quad (3.75)$$

where $A[k]$ is the frequency domain acceleration content at frequency index k , $a[n]$ is the acceleration sample at time index n and $e^{-j2\pi kn/N}$ is the complex sinusoidal basis function.

The physical frequency corresponding to the k^{th} FFT bin is,

$$f_k = \frac{k}{N} f_s \quad (3.76)$$

where f_s is the sampling frequency.

The power contained in each frequency component is proportional to the squared magnitude of the Fourier coefficient,

$$\text{Power} \propto |A[k]|^2 \quad (3.77)$$

A simple periodogram estimate of the acceleration PSD can therefore be written as,

$$S_{aa}(f_k) = \frac{|A[k]|^2}{N f_s} \quad (3.78)$$

where $S_{aa}(f)$ is the acceleration power spectral density with units of,

$$\frac{(m/s^2)^2}{Hz} \quad (3.79)$$

In continuous form, the PSD may be interpreted as,

$$S_{aa}(f) = \lim_{T \rightarrow \infty} \frac{1}{T} |A(f)|^2 \quad (3.80)$$

This means that the PSD represents how much acceleration variance is present at each frequency.

3.6.3 Manual Implementation of Welch's Method in MATLAB

A single FFT based periodogram can be noisy especially for road vibration signals because vehicle vibration contains random excitation, suspension response, impact events and measurement noise. Therefore, Welch's method was used to obtain a smoother PSD estimate.

Welch's method was implemented manually using basic MATLAB operations such as signal segmentation, windowing, FFT calculation, magnitude-squared spectrum calculation and averaging.

In the manual Welch implementation, the acceleration signal was divided into shorter overlapping segments. If the segment length is N_w , the number of overlapping samples is N_{overlap} and the step size between two consecutive segments is D , then,

$$D = N_w - N_{\text{overlap}} \quad (3.81)$$

The number of segments is,

$$L = \left\lfloor \frac{N - N_{\text{overlap}}}{D} \right\rfloor \quad (3.82)$$

For the r^{th} segment, the selected acceleration samples are,

$$a_r[n] = a[n + rD] \quad (3.83)$$

A window function was applied to each segment to reduce spectral leakage. For a general window $w[n]$, the windowed segment becomes,

$$a_{rw}[n] = w[n]a_r[n] \quad (3.84)$$

The Welch PSD estimate is then obtained by averaging the periodograms from all segments:

$$S_{aa}(f_k) = \frac{1}{L} \sum_{r=1}^L P_r(f_k) \quad (3.85)$$

This averaging process reduces random fluctuations in the PSD and gives a smoother estimate of the road vibration frequency content.

To further improve smoothness, the overlap percentage can be increased. In this project, the PSD smoothness was improved by using overlapping segments. A higher overlap, such as approximately 75%, increases the number of averaged segments and reduces spectral variation. However, the segment length must be selected carefully because a shorter segment gives a smoother PSD but reduces frequency resolution.

The PSD was also represented in decibel scale using,

$$PSD_{dB} = 10 \log_{10}(S_{aa}(f)) \quad (3.86)$$

If,

$$S_{aa}(f) = 0.431325 \quad (3.87)$$

Then,

$$PSD_{dB} = 10 \log_{10}(0.431325) = -3.65196 \text{ dB} \quad (3.88)$$

3.6.4 Road PSD-Based Power Estimation

From the non-resonant analytical model, the frequency response denominator of the harvester can be written as,

$$H(\Omega) = H_R(\Omega) + jH_I(\Omega) \quad (3.89)$$

where the real part is,

$$H_R(\Omega) = 1 - \Omega^2 + \frac{k^2 \Omega^2}{\Omega^2 + 4(\xi_e + \xi_c)^2} \quad (3.90)$$

and the imaginary part is,

$$H_I(\Omega) = \frac{\Omega}{Q_m} + \frac{2k^2(\xi_e + \xi_c)\Omega}{\Omega^2 + 4(\xi_e + \xi_c)^2} \quad (3.91)$$

So,

$$|H(\Omega)| = \sqrt{\left[1 - \Omega^2 + \frac{k^2 \Omega^2}{\Omega^2 + 4(\xi_e + \xi_c)^2}\right]^2 + \left[\frac{\Omega}{Q_m} + \frac{2k^2(\xi_e + \xi_c)\Omega}{\Omega^2 + 4(\xi_e + \xi_c)^2}\right]^2} \quad (3.92)$$

For a harmonic acceleration component, the harvested power across the load can be written as,

$$P_R = k^2 \xi_c M \omega_0^3 \left[\frac{\Omega^2}{\Omega^2 + 4(\xi_e + \xi_c)^2} \right] u_m^2 \quad (3.93)$$

Substituting,

$$u_m^2 = \frac{\gamma_m^2}{\omega_0^4 |H(\Omega)|^2} \quad (3.94)$$

Gives,

$$P_R = \frac{k^2 \xi_c M}{\omega_0} \left[\frac{\Omega^2}{\Omega^2 + 4(\xi_e + \xi_c)^2} \right] \frac{\gamma_m^2}{|H(\Omega)|^2} \quad (3.95)$$

For random road vibration, the term γ_m^2 is replaced by the acceleration variance contribution at each frequency,

$$\gamma_m^2 \rightarrow S_{aa}(f) df \quad (3.96)$$

Therefore, the total expected harvested power from the measured road acceleration PSD is,

$$P_{\text{total}} = \int_0^\infty \left[\frac{k^2 \xi_c M}{\omega_0} \left(\frac{\Omega^2}{\Omega^2 + 4(\xi_e + \xi_c)^2} \right) \frac{1}{|H(\Omega)|^2} \right] S_{aa}(f) df \quad (3.97)$$

Where,

$$\Omega = \frac{2\pi f}{\omega_0} \quad (3.98)$$

Above equation combines the measured road vibration PSD with the analytical transfer function of the electromagnetic harvester. Therefore, the power estimate is not based only on the size of the road vibration peaks, but also on how strongly the harvester responds at each frequency.

3.6.5 Interpretation of PSD Results

The Welch PSD plots were used to identify the frequency bands where the road-induced acceleration energy was concentrated. A peak in $S_{aa}(f)$ indicates that a large portion of the acceleration variance exists near that frequency. However, the useful energy for harvesting also depends on the harvester transfer function. A frequency band with moderate acceleration PSD can still produce significant output if it is close to the harvester natural frequency.

The PSD analysis confirmed that the lower control arm vibration is broadband and road-dependent. However, the road tests showed repeated low-frequency vibration content for different road conditions. Therefore, the PSD results supported the design decision to tune the harvester near the dominant frequency range identified from the road vibration data.

The calculated road acceleration PSD was further used to estimate the harvester displacement PSD and expected total output power using the analytical transfer function of the electromagnetic vibration energy harvester.

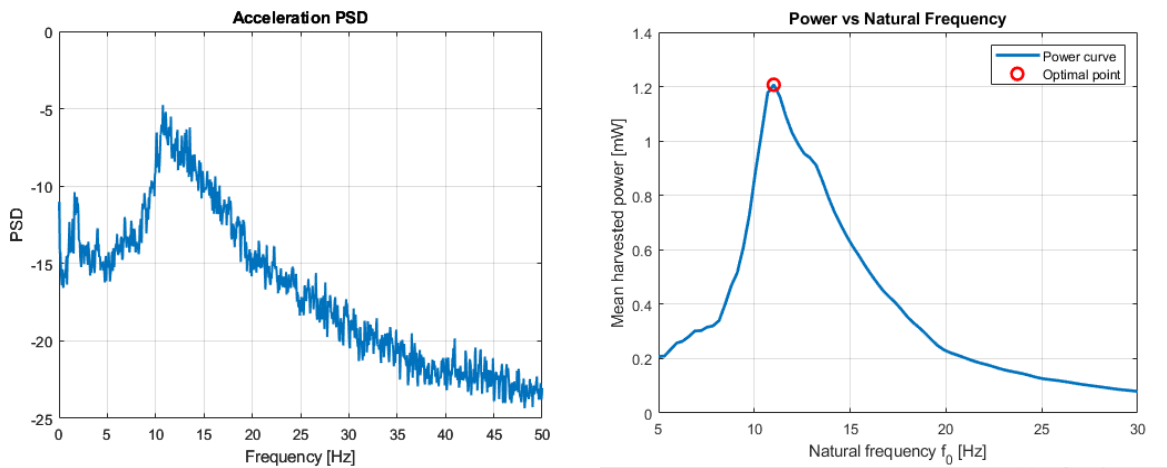


Figure 3.4: Highway (E road) road profile

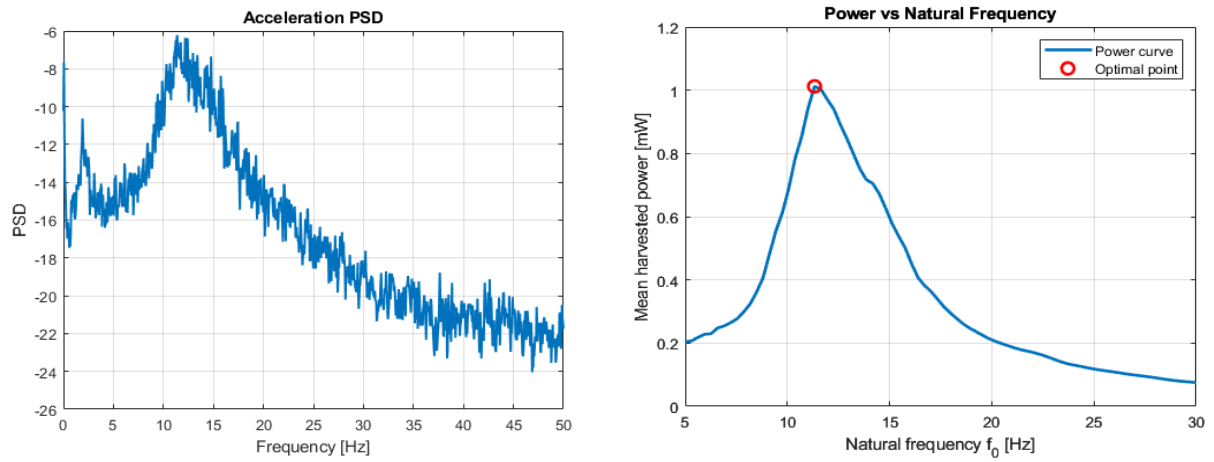


Figure 3.5: Highway (E road) road second profile

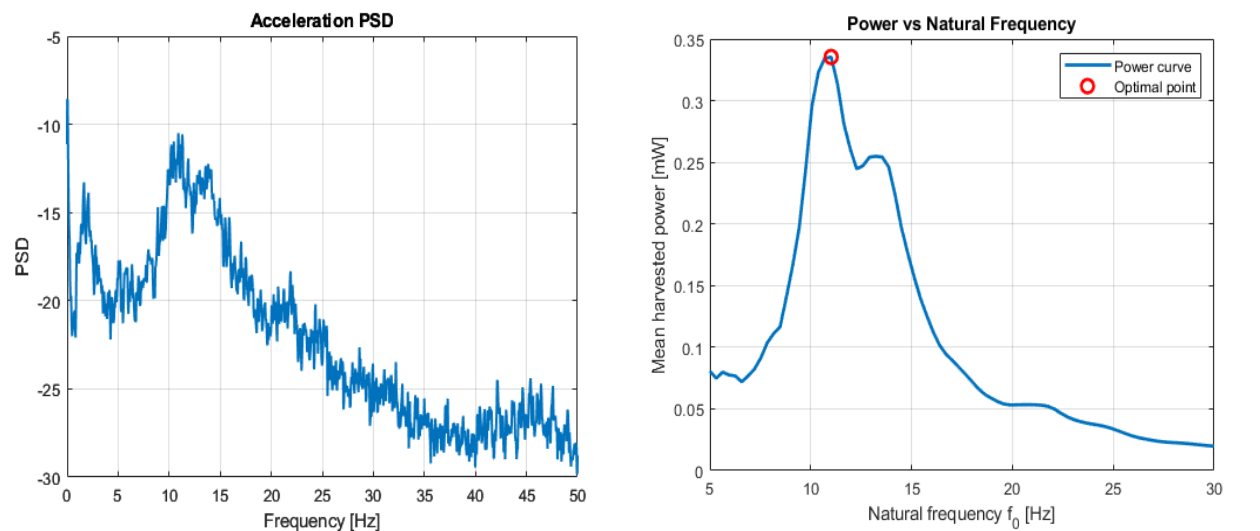


Figure 3.6: A road profile

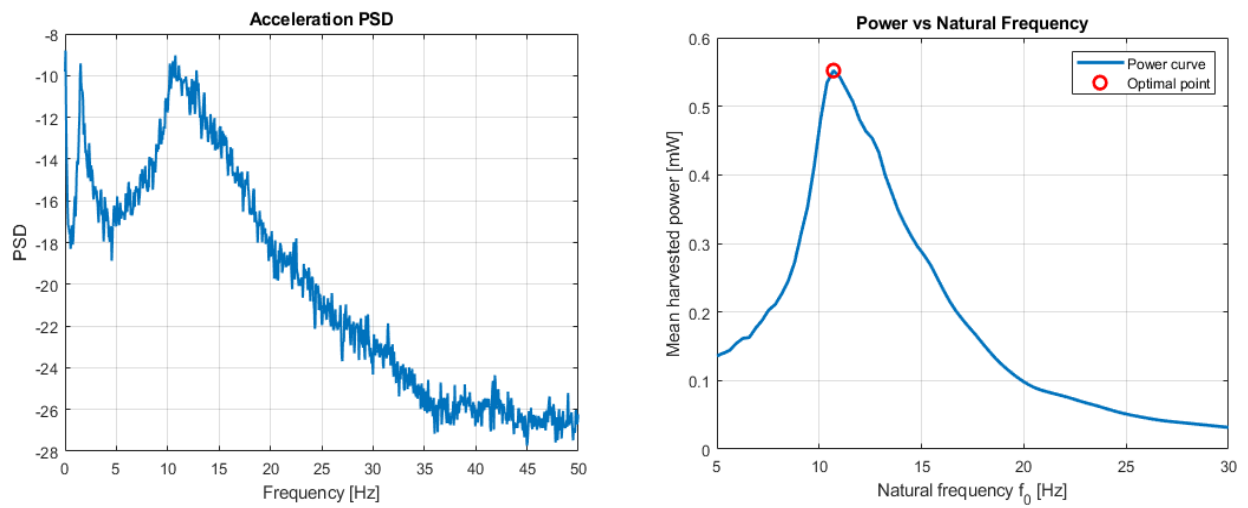


Figure 3.7: A road second profile

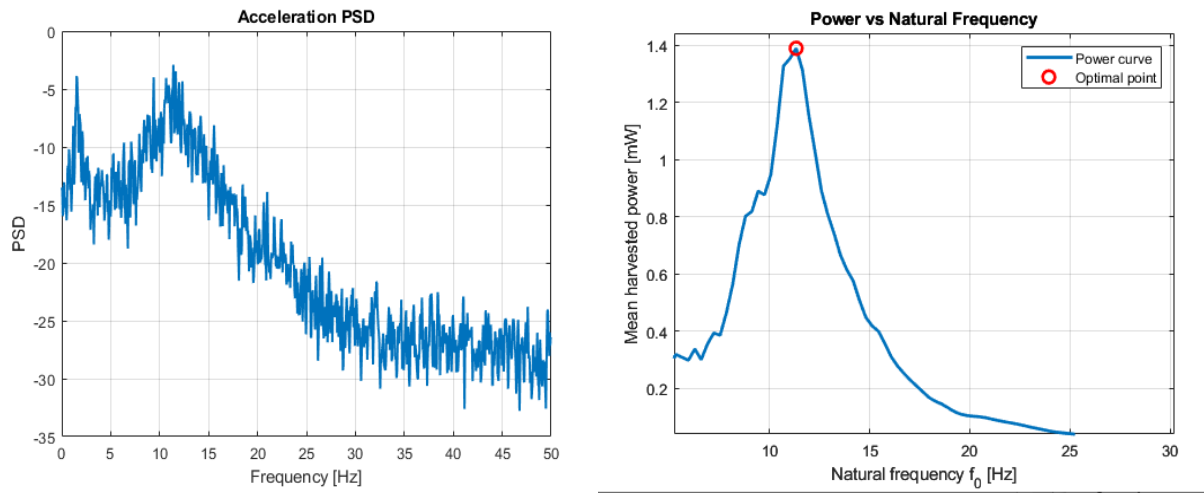


Figure 3.11: Concrete road second profile

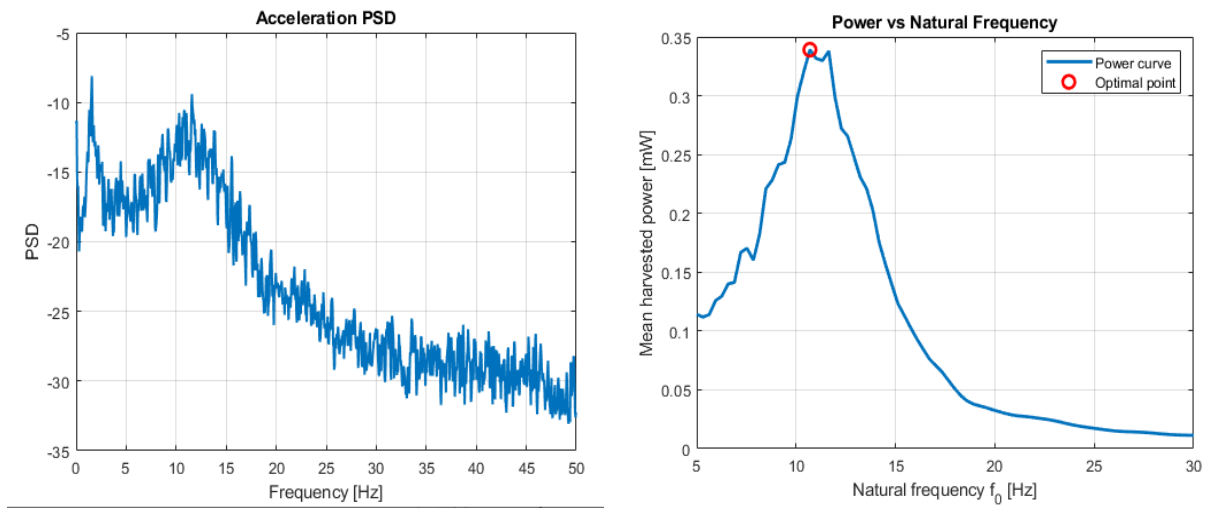


Figure 3.9: B road second profile

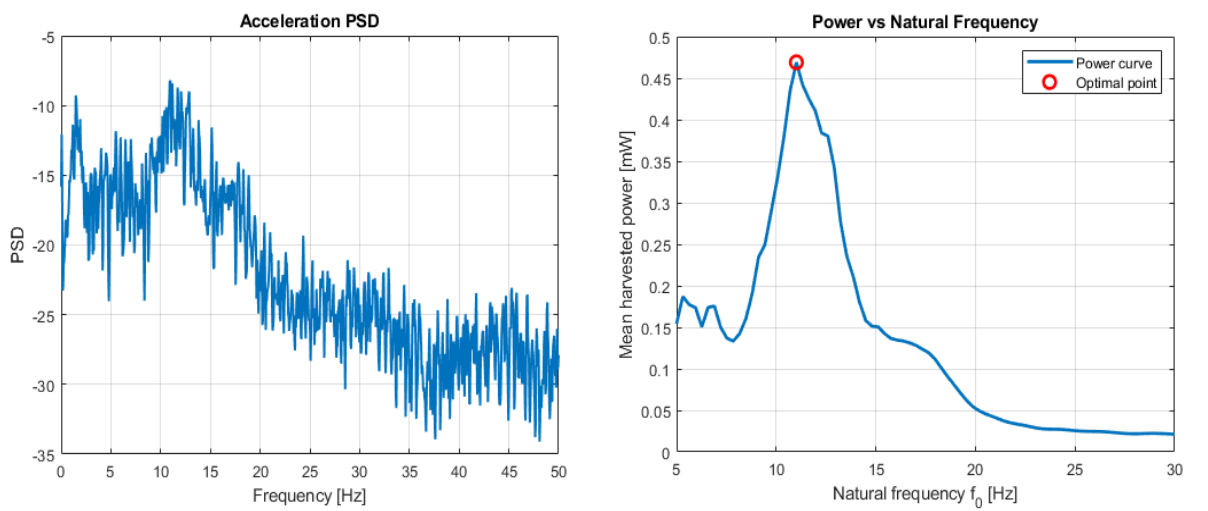


Figure 3.8: Concrete road profile

3.7 Numerical and Parametric Analysis

The analytical equations were implemented in MATLAB to evaluate the behaviour of the harvester under different design conditions. The main dummy input parameters used in the model are summarized in Table 3.5.

Table 3.5: Input parameters used for the analytical model

Parameter	Symbol	Value
Seismic mass	M	0.2 kg
Target natural frequency	f_0	11 Hz
Mechanical quality factor	Q_m	30
Damping ratio	ζ	0.0167
Coupling product	$k^2 Q_m$	5
Squared coupling coefficient	k^2	0.1667
Resistive loss coefficient	ξ_e	1.5

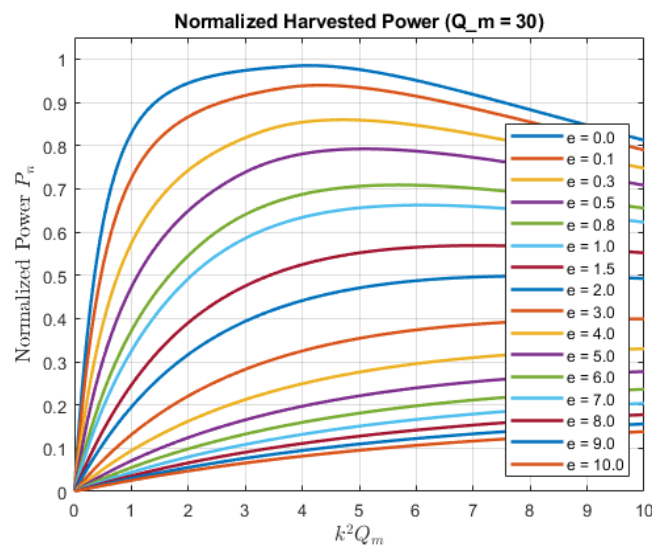


Figure 3.12: Normalized harvested power Vs $k^2 Q_m$

The graph shows that normalized harvested power increases with $k^2 Q_m$ at first because stronger electromechanical coupling improves energy conversion. However, at higher $k^2 Q_m$, the power can reduce due to increased electrical damping, which limits the relative motion of the seismic mass. The results also show that higher ξ_e significantly reduces the maximum power because of increased resistive losses. Therefore, good harvester performance requires a suitable coupling level and low coil resistance.

3.7.1 Effect of excitation frequency

The output is highest when the excitation frequency is close to the natural frequency of the harvester. Therefore, the spring stiffness must be selected according to the dominant vibration frequency obtained from the vehicle test data.

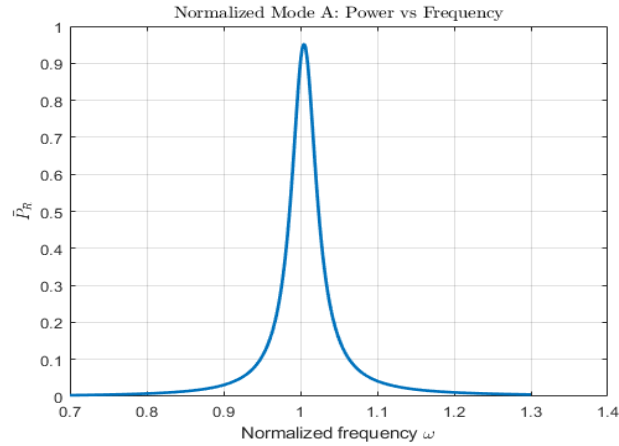
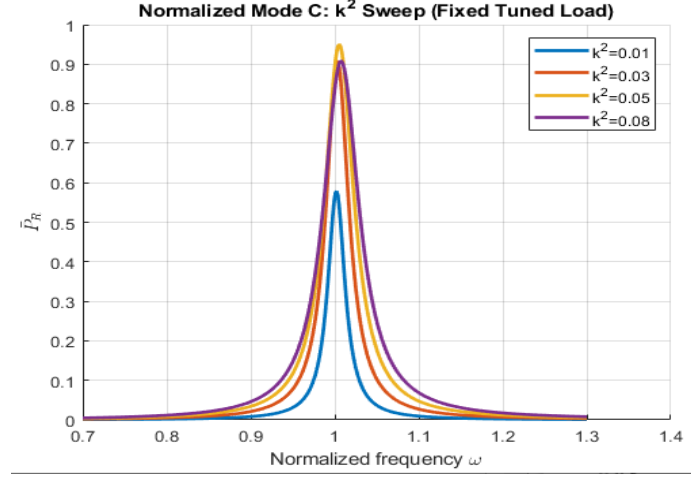


Figure 3.13: Normalized power Vs Normalized frequency

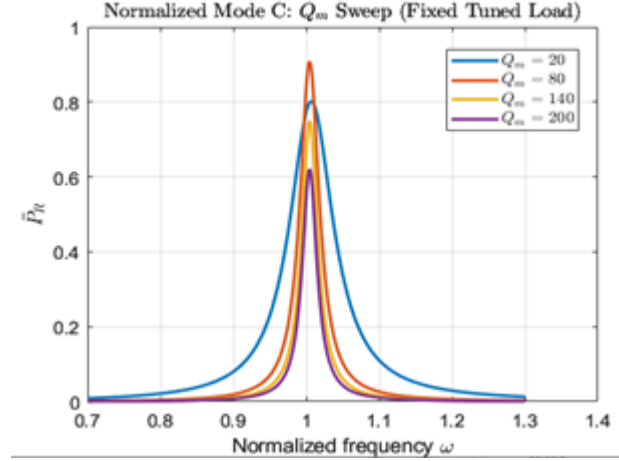
3.7.2 Effect of Squared Electromechanical Coupling Coefficient

The figure 3.14 shows that normalized harvested power peaks near $\Omega = 1$, confirming maximum output at resonance. Increasing k^2 improves power up to about $k^2 = 0.05$, but further increase reduces the peak slightly due to higher electromagnetic damping.

Figure 3.14: k^2 Vs Normalized frequency

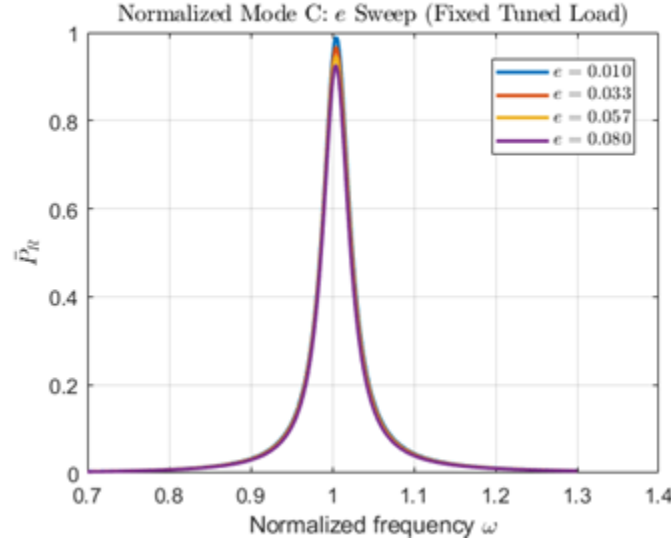
3.7.3 Effect of Mechanical Quality Factor

The figure 3.15 shows that normalized harvested power peaks near $\Omega = 1$, confirming best performance at resonance. Increasing Q_m improves power up to about $Q_m = 80$, but higher values reduce the peak and narrow the response. Therefore, a moderate Q_m gives the best balance between power output and bandwidth.

Figure 3.15: Q_m Vs Normalized frequency

3.7.4 Effect of Resistive Loss Coefficient

The figure 3.16 shows that normalized harvested power peaks near $\Omega = 1$ for all ξ_e values. As ξ_e increases, the peak power slightly decreases due to higher coil resistance losses. Although the effect is small in the selected range, reducing ξ_e improves the useful power delivered to the load.

Figure 3.16: ξ_e Vs Normalized frequency

3.8 Design and Parameter Selection

3.8.1 Introduction

This section presents the design development and parameter selection process of the proposed electromagnetic vibration energy harvester. The purpose of this stage was to convert the selected harvesting concept into a practical physical design that can be fabricated and tested. The design was developed based on vehicle vibration characteristics, electromagnetic energy harvesting principles, manufacturability, material availability and compatibility with the GPS tracking unit.

The selected system is a linear electromagnetic vibration energy harvester. It operates by allowing relative motion between a permanent magnet and a copper coil. When the vehicle vibrates, the moving magnetic element oscillates relative to the coil causing a change in magnetic flux linkage. This produces an induced voltage according to Faraday's law of electromagnetic induction. The main design parameters considered in this section include the dynamic mass, spring stiffness, magnet type, coil design, housing material and final system configuration.

3.8.2 Design Requirements

The energy harvester was designed to operate under vehicle induced vibration conditions and provide supplementary electrical energy to support a discrete vehicle tracking unit. Therefore,

the design requirements were defined based on functional performance, mechanical feasibility, electromagnetic performance and prototype manufacturability.

Table 3.6: Main design requirements

Requirement	Description
Energy harvesting mechanism	The system should convert vehicle vibration into electrical energy.
Operating vibration direction	The design should mainly respond to vertical vibration since vertical acceleration was identified as the dominant excitation direction.
Compactness	The harvester should be small enough to be installed inside or near a vehicle mounted tracking system.
Durability	The design should withstand repeated vibration during vehicle operation.
Low friction motion	The moving magnetic element should move freely with minimum mechanical resistance.
Manufacturability	Components should be suitable for prototype fabrication using available manufacturing methods.
Non magnetic housing	The housing material should not significantly disturb the magnetic field.
Electrical output	The system should generate measurable induced voltage under vibration excitation.
System compatibility	The harvester should be suitable for future integration with rectification, power management, battery storage and GPS tracker operation.

3.8.3 Conceptual Design

The conceptual design was developed based on a linear electromagnetic energy harvesting arrangement. The system consists of a moving magnet assembly, a stationary or relatively fixed

coil, compression springs, a guide shaft and an outer housing. When the vehicle experiences vibration, the magnet assembly moves relative to the coil. This relative motion changes the magnetic flux linkage through the coil and generates an induced voltage.

Several design concepts were initially considered.

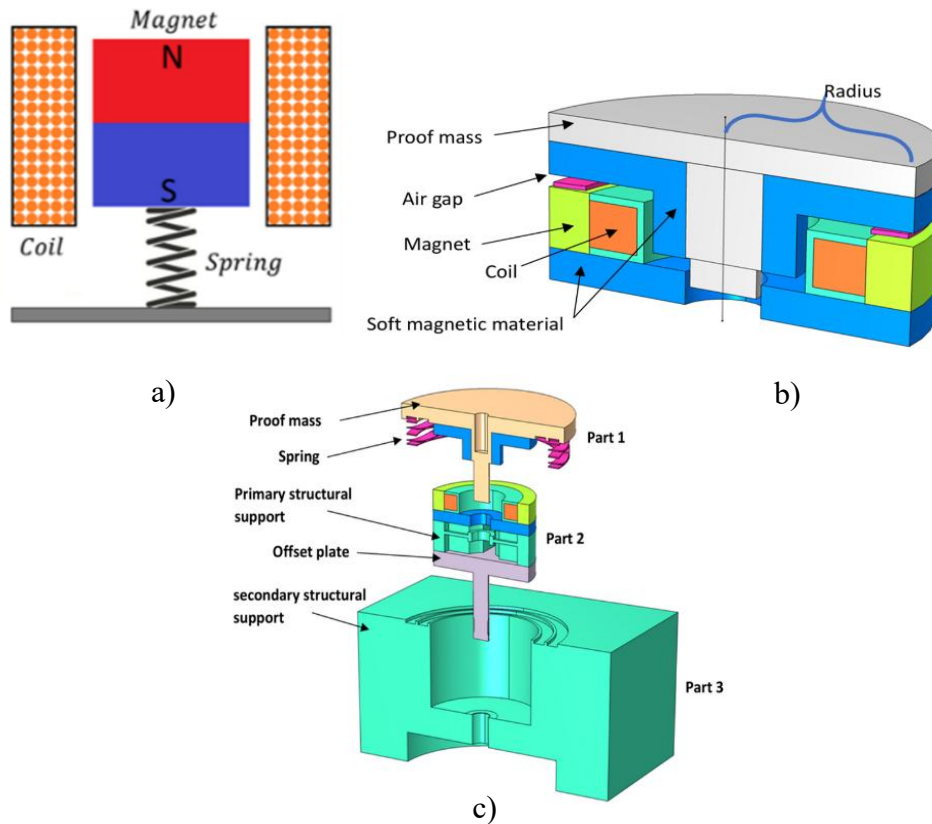


Figure 3.17: Electromagnetic energy harvester configurations considered during concept selection:
 (a) linear magnet–coil spring-based harvester, (b) rotational electromagnetic harvester configuration
 (c) spring-supported multi-part electromagnetic harvester assembly

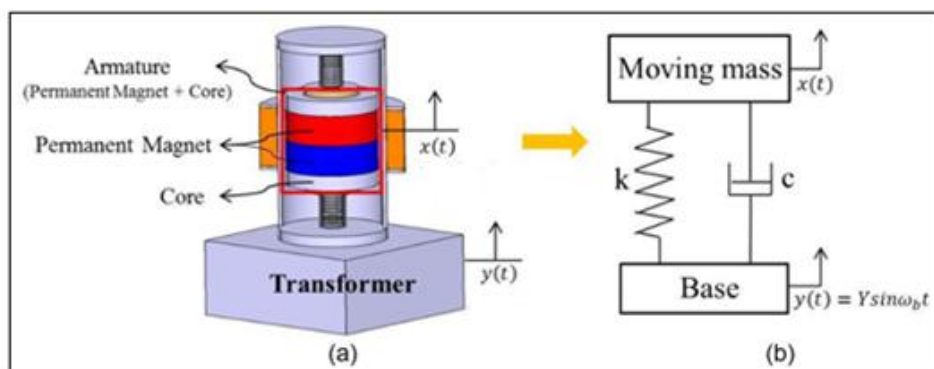


Figure 3.18: Selected concept design

The selected concept was chosen because it is simple, compact, and suitable for prototype fabrication. The final concept uses a cylindrical arrangement, which also supports axisymmetric electromagnetic simulation in COMSOL. This is useful because the same physical design can be simplified into a 2D axisymmetric model for simulation reducing computational time while preserving the main electromagnetic behaviour.

The main reasons for selecting the linear electromagnetic concept are:

Table 3.7 - Key design considerations for selecting the linear electromagnetic energy harvester concept

Design Consideration	Reason
Simplicity	Easier to design, model, fabricate, and test.
Electromagnetic induction	Provides direct voltage generation from magnet-coil relative motion.
Prototype feasibility	Can be built using common components such as magnets, copper coil, springs, shaft, and 3D printed housing.
Simulation compatibility	Cylindrical geometry allows 2D axisymmetric modelling.

The conceptual assembly developed in SolidWorks included the armature, casing, copper coil, compression springs, ring-type cylindrical magnet, shaft, and optional core.

After selecting the linear electromagnetic concept, the initial hand sketch was converted into a 3D CAD model using SolidWorks, as shown in Fig. 3.19 and Fig. 3.20. The hand sketch was used to define the basic component arrangement while the CAD model was used to develop the final geometry for assembly, simulation and prototype fabrication.

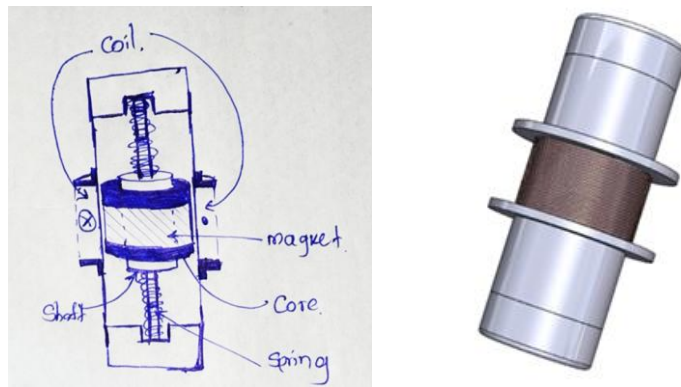


Figure 3.19: a) Initial Hand sketch

b) Initial SolidWorks CAD model of the selected electromagnetic harvester concept

3.8.4 Main Components

The final harvester design consists of several main mechanical and electromagnetic components. Each component performs a specific function in the energy conversion process.

Table 3.8: Components of the harvester and their functions

Component	Function
Permanent magnet	Produces the magnetic field required for electromagnetic induction.
Copper coil	Generates induced voltage when magnetic flux linkage changes.
Compression springs	Provide restoring force and allow oscillatory motion.
Proof mass / moving assembly	Responds to vehicle vibration and creates relative motion.
Shaft	Guides the moving assembly and maintains linear motion.
Housing / casing	Supports and protects the internal components.
Core / core-less region	Influences magnetic flux concentration and mechanical motion.

The selected permanent magnet and moving assembly act as the oscillating element. The compression springs restore the magnet assembly after displacement and allow repeated vibration response. The copper coil is positioned so that the magnetic flux through the coil changes as the magnet moves. The shaft maintains the alignment of the moving element and prevents unwanted lateral motion.

The housing provides structural support and protects the coil, magnet and spring arrangement. PLA was selected as the intended housing material for prototype fabrication because it is lightweight, non-magnetic and suitable for 3D printing. The annexed component list identifies the main materials used for the harvester as NdFeB magnet, steel core, copper coil and PLA casing.

The internal arrangement of the harvester is shown in Fig. 3.21. The ring-type cylindrical magnet and core are placed along the central shaft while the copper coil is positioned around the magnet region. The compression springs are located at both ends of the moving assembly to provide restoring force during vibration. This arrangement allows the magnet assembly to move linearly relative to the coil producing a changing magnetic flux linkage required for electromagnetic induction.

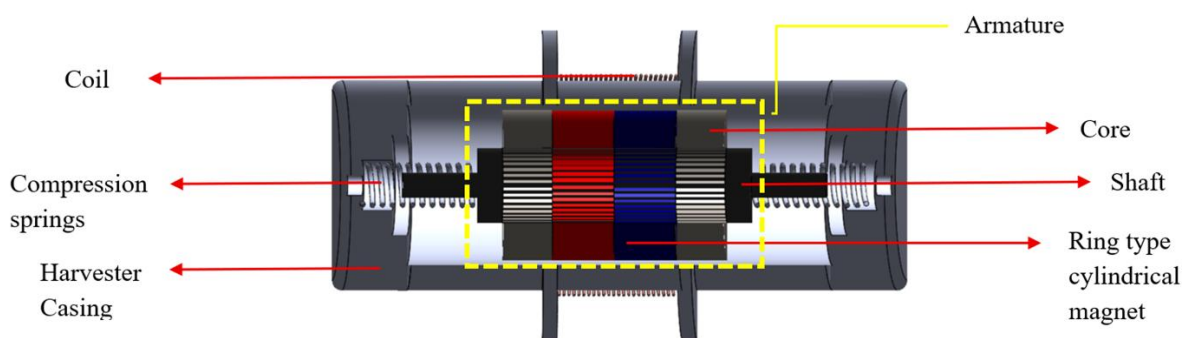


Figure 3.20 - Section view of the electromagnetic vibration energy harvester showing the internal component arrangement

The exploded view of the harvester assembly is shown in Fig. 3.22. This view shows the arrangement of the individual components before assembly including the housing, coil, springs, shaft, core, magnet and end supports.

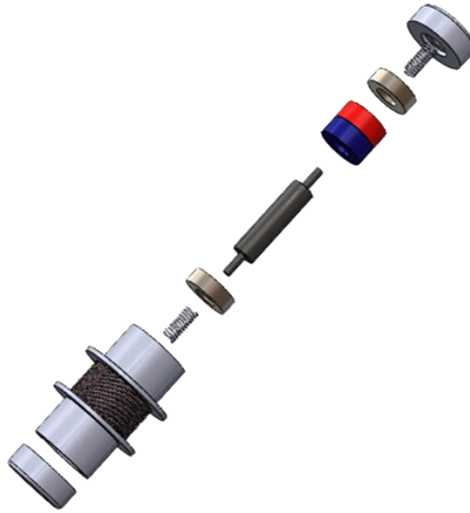


Figure 3.21 - Exploded view of the electromagnetic vibration energy harvester assembly

3.8.5 Mathematical Structure for Parameter Determination

Table 3.9: Equation parameters

Parameter	Symbol	Source
Target natural frequency	f_0	PSD analysis of the road vibration
Seismic mass	M	Selected
Number of coil turns	N	Known from coil winding
Coil resistance	r_L	Measured using multimeter
Coil inductance	L_0	Measured using LCR meter
Magnetic flux density profile	$B(z)$	Measured using Gauss meter
Mean coil diameter	D_{mean}	Measured from coil geometry

3.8.5.1 Proof Mass and Spring Stiffness

The natural angular frequency is,

$$\omega_0 = 2\pi f_0 \quad (3.99)$$

The proof mass is one of the main parameters that controls the dynamic response of the vibration energy harvester. In this design, the magnet assembly is considered as part of the proof mass. This is useful because the magnet performs two functions at the same time. It provides the magnetic field required for electromagnetic induction and also contributes to the moving mass of the oscillator. Additional mass can be added to the moving assembly if resonance tuning is required after fabrication.

The target natural frequency was selected from the road vibration analysis. Once the target frequency f_0 and proof mass M are selected, the required spring stiffness is calculated from,

$$\begin{aligned} \omega_0 &= 2\pi f_0 \\ K &= M\omega_0^2 \end{aligned} \quad (3.100)$$

Therefore, the selected mass directly affects the required spring stiffness. A larger mass reduces the required stiffness for the same natural frequency while a smaller mass requires a stiffer spring system.

The selected design procedure is,

$$M \rightarrow f_0 \rightarrow \omega_0 \rightarrow K$$

This means that the proof mass is selected first based on mechanical feasibility and then the required spring stiffness is calculated to tune the harvester to the target road vibration frequency.

The required equivalent spring stiffness is,

$$K = M(2\pi f_0)^2 \quad (3.101)$$

where M is the moving mass and f_0 is the target natural frequency.

If two identical springs are used in parallel,

$$K_{eq} = K_1 + K_2 \quad (3.102)$$

$$K_s = \frac{K_{eq}}{2} \quad (3.103)$$

where K_s is the stiffness of each spring.

The final spring selection must also consider the maximum allowable displacement, available space, spring preload and avoidance of collision between the magnet and the coil. Since the actual spring stiffness can vary from the catalogue value, the final resonance frequency should be checked experimentally. If the measured natural frequency is higher or lower than the target value, it can be corrected by changing the spring stiffness or adjusting the moving mass.

3.8.5.2 Mechanical Damping

The mechanical quality factor is,

$$Q_m = \frac{M\omega_0}{D} \quad (3.104)$$

Therefore, the mechanical damping coefficient is,

$$D = \frac{M\omega_0}{Q_m} \quad (3.105)$$

The damping ratio is,

$$\zeta = \frac{1}{2Q_m} \quad (3.106)$$

3.8.5.3 Magnet Selection

The permanent magnet is the source of magnetic flux in the electromagnetic vibration energy harvester. A neodymium magnet was selected because it can produce a strong magnetic field in a compact volume.

In the design procedure, the magnetic field of the actual magnet was measured using a Gauss meter. This gives a more realistic estimate of the magnetic flux density than using only catalogue data. This measured-field method is more accurate than assuming an ideal magnet because it uses the actual magnetic behaviour of the selected magnet. The obtained β_{rms} is then used directly to calculate the dimensionless coupling coefficient k^2 .

Magnetic flux linkage from Gaus meter data

The Gauss meter gives the magnetic flux density along the direction of magnet motion. The measured field is written as $B_z(z)$.

If the magnetic field is measured in millitesla, it is first converted to tesla,

$$B_z(T) = B_z(mT) \times 10^{-3} \quad (3.107)$$

The effective coil area is,

$$A_{\text{eff}} = \pi r_{\text{mean}}^2 \quad (3.108)$$

Where,

$$r_{\text{mean}} = \frac{D_{\text{mean}}}{2} \quad (3.109)$$

The magnetic flux through one turn is,

$$\Phi(z) = B_z(z)A_{\text{eff}} \quad (3.110)$$

The flux linkage of the full coil is,

$$\lambda(z) = N\Phi(z) = NA_{\text{eff}}B_z(z) \quad (3.111)$$

If the RMS magnetic flux density is used,

$$B_{\text{rms}} = \sqrt{\frac{1}{z_2 - z_1} \int_{z_1}^{z_2} B_z^2(z) dz} \quad (3.112)$$

then the RMS flux linkage can be written as,

$$\lambda_{\text{rms}} = NA_{\text{eff}}B_{\text{rms}} \quad (3.113)$$

However, for voltage and power prediction, the important parameter is not only B_{rms} but the change of flux linkage with displacement.

Electromagnetic coupling coefficient

The electromagnetic coupling coefficient is,

$$\beta(z) = \frac{d\lambda(z)}{dz} \quad (3.114)$$

Since,

$$\lambda(z) = NA_{\text{eff}}B_z(z) \quad (3.115)$$

Then,

$$\beta(z) = NA_{\text{eff}} \frac{dB_z(z)}{dz} \quad (3.116)$$

The RMS coupling coefficient is,

$$\beta_{\text{rms}} = \sqrt{\frac{1}{z_2 - z_1} \int_{z_1}^{z_2} \beta^2(z) dz} \quad (3.117)$$

For discrete Gauss meter readings,

$$\beta_{\text{rms}} = \sqrt{\frac{\sum \beta^2(z_i) \Delta z_i}{z_2 - z_1}} \quad (3.118)$$

This is the way of obtaining β because it uses the measured magnetic field of the actual magnet.

3.8.5.4 Dimensionless Parameters Assumption

In the initial design stage, coil parameters such as resistance, inductance, wire length and number of turns were estimated from assumed geometry and wire diameter. Therefore, the dimensionless parameters are calculated directly.

The squared electromechanical coupling coefficient is,

$$k^2 = \frac{\beta_{\text{rms}}^2}{KL_0} \quad (3.119)$$

The resistive loss coefficient is,

$$\xi_e = \frac{r_L}{2\omega_0 L_0} \quad (3.120)$$

The load coefficient is,

$$\xi_c = \frac{R}{2\omega_0 L_0} \quad (3.121)$$

The coupling-quality factor product is $k^2 Q_m$.

This value is useful for comparing the design with the normalized harvested power curves.

3.8.5.5 Load Tuning

For a given k^2 , Q_m and ξ_e , the load coefficient ξ_c can be swept to find the optimum load.

The denominator used in the normalized non-resonant model is,

$$D_e(\Omega) = \Omega^2 + 4(\xi_c + \xi_e)^2 \quad (3.122)$$

$$D_R(\Omega) = 1 - \Omega^2 + k^2 \left(\frac{\Omega^2}{D_e(\Omega)} \right) \quad (3.123)$$

$$D_I(\Omega) = \frac{\Omega}{Q_m} + k^2 \left(\frac{2\Omega(\xi_c + \xi_e)}{D_e(\Omega)} \right) \quad (3.124)$$

$$|D(\Omega)|^2 = D_R^2(\Omega) + D_I^2(\Omega) \quad (3.125)$$

The normalized harvested power is,

$$P_n = \frac{8k^2 \xi_c}{Q_m} \left(\frac{\Omega^2}{D_e(\Omega)} \right) \frac{1}{|D(\Omega)|^2} \quad (3.126)$$

At resonance $\Omega = 1$.

The optimum load coefficient is found by,

$$\xi_{c,\text{opt}} = \arg \max_{\xi_c} (P_n) \quad (3.127)$$

Then the optimum physical load resistance is,

$$R_{\text{opt}} = 2\omega_0 L_0 \xi_{c,\text{opt}} \quad (3.128)$$

3.8.5.6 Coil Design

The coil is the electrical component in which voltage is induced. In the initial design stage, coil parameters such as resistance, inductance, wire length and number of turns were estimated from assumed geometry and wire diameter. However, since the coil has now been fabricated, the updated method uses measured coil parameters.

Table 3.10: Coil design parameters

Parameter	Symbol
Number of turns	N
Coil resistance	r_L
Coil inductance	L_0
Mean coil diameter	D_{mean}
Effective coil area	A_{eff}

The coil resistance r_L can be measured using a multimeter and the inductance L_0 can be measured using an LCR meter. These measured values are used directly in the dimensionless parameter calculation.

The induced voltage is related to the relative velocity by,

$$v_{\text{oc}}(t) = \beta \dot{u}(t) \quad (3.129)$$

where β is calculated from the measured magnetic field and known coil turns.

The squared electromechanical coupling coefficient is,

$$k^2 = \frac{\beta_{\text{rms}}^2}{KL_0} \quad (3.130)$$

The resistive loss coefficient is,

$$\xi_e = \frac{r_L}{2\omega_0 L_0} \quad (3.131)$$

The actual resistance and inductance of the fabricated coil are used instead of theoretical estimates. It also allows the analytical model to be updated after coil fabrication.

3.8.5.7 Dimensionless Parameter Determination

Since the coil is already made, L_0 , r_L and N are known. After selecting the proof mass, spring stiffness, magnet and coil, the system can be described using dimensionless parameters. These parameters are useful because they combine the mechanical and electrical behaviour of the harvester into a compact form.

The mechanical quality factor is,

$$Q_m = \frac{M\omega_0}{D} \quad (3.132)$$

If Q_m is selected or experimentally estimated, the damping coefficient becomes,

$$D = \frac{M\omega_0}{Q_m} \quad (3.133)$$

The squared electromechanical coupling coefficient is,

$$k^2 = \frac{\beta_{\text{rms}}^2}{KL_0} \quad (3.134)$$

The resistive loss coefficient is,

$$\xi_e = \frac{r_L}{2\omega_0 L_0} \quad (3.135)$$

The load coefficient is,

$$\xi_c = \frac{R}{2\omega_0 L_0} \quad (3.136)$$

These parameters are calculated using the actual coil resistance, coil inductance, coil turns and measured magnetic field data. Therefore, the updated parameter determination procedure is,

$$\begin{aligned}
 B_z(z), N, A_{\text{eff}} &\rightarrow \beta_{\text{rms}} \\
 M, f_0 &\rightarrow K, \omega_0 \\
 \beta_{\text{rms}}, K, L_0 &\rightarrow k^2 \\
 r_L, \omega_0, L_0 &\rightarrow \xi_e \\
 R, \omega_0, L_0 &\rightarrow \xi_c
 \end{aligned}$$

This procedure reduces uncertainty because fewer values are assumed. The only major parameters that still need tuning are the mechanical quality factor Q_m and the external load resistance R .

3.8.5.8 Load Resistance Selection

The load resistance affects the amount of electrical power that can be extracted from the coil. If the load resistance is too low, the output voltage becomes small. If the load resistance is too high, the current becomes small. Therefore, an optimum load resistance exists.

Instead of selecting the load directly in ohms at first, the normalized load coefficient is used,

$$\xi_c = \frac{R}{2\omega_0 L_0} \quad (3.137)$$

The load coefficient is swept in MATLAB and the value that gives the maximum normalized harvested power is selected as,

$$\xi_{c,\text{opt}} = \arg \max_{\xi_c} (P_n) \quad (3.138)$$

The corresponding physical load resistance is then calculated using,

$$R_{\text{opt}} = 2\omega_0 L_0 \xi_{c,\text{opt}} \quad (3.139)$$

This method is useful because it uses the measured inductance of the actual coil. After fabrication, the theoretical optimum load should be verified experimentally by connecting different load resistors and measuring the output voltage and power.

3.9 COMSOL Simulation and Numerical Analysis

3.9.1 Introduction

This chapter presents the COMSOL Multiphysics simulation and numerical analysis carried out for the proposed electromagnetic vibration energy harvester. The main purpose of the simulation was to evaluate the electromagnetic behaviour of the selected harvester design before prototype fabrication and experimental testing. The simulation was used to analyse magnetic flux distribution, flux linkage variation through the coil, and the induced voltage generated due to relative motion between the magnet and coil.

A 2D axisymmetric modelling approach was used because the harvester has a cylindrical geometry. This approach reduced computational time while still representing the main electromagnetic behaviour of the system. The simulation study was used to compare different design configurations, including casing and no-casing arrangements, as well as core and core-less configurations. The results were then used to support design decisions and identify the most suitable configuration for the initial prototype.

The COMSOL simulation also provided a numerical basis for future optimisation of the harvester. Key design parameters such as magnet position, air gap, coil arrangement, casing material, and vibration input can be modified in the model to study their effect on voltage generation and overall energy harvesting performance.

3.9.2 Simulation Geometry

The electromagnetic harvester was first developed as a 3D CAD model using SolidWorks. Since the design has a cylindrical arrangement, the COMSOL model was simplified into a 2D axisymmetric geometry. This simplification was suitable because the main components, including the ring-type magnet, coil, shaft region, and surrounding air domain, are arranged around a central axis.

The 2D axisymmetric model included the main electromagnetic regions required for simulation. These included the permanent magnet, copper coil region, air domain, optional casing, and optional ferromagnetic core. The air domain was included around the magnet and coil to allow the magnetic field to develop properly and to reduce boundary effects.

The geometry was created to represent the relative motion between the magnet and coil. A parametric displacement was applied to the magnet position so that the change in magnetic flux linkage could be observed at different locations. The magnet position varied from -20 mm to +20 mm during the parametric sweep study.

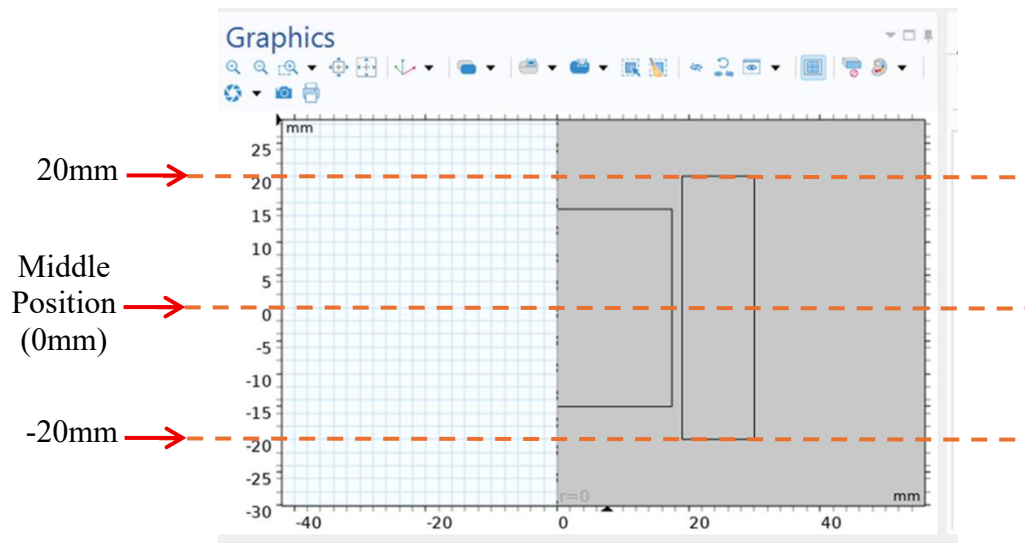


Figure 3.22: 2D axisymmetric geometry of the electromagnetic vibration energy harvester

The 2D axisymmetric geometry used in the COMSOL simulation is shown in Fig. 3.23. The model includes the permanent magnet, copper coil region, air domain, and optional casing arranged around the central axis.

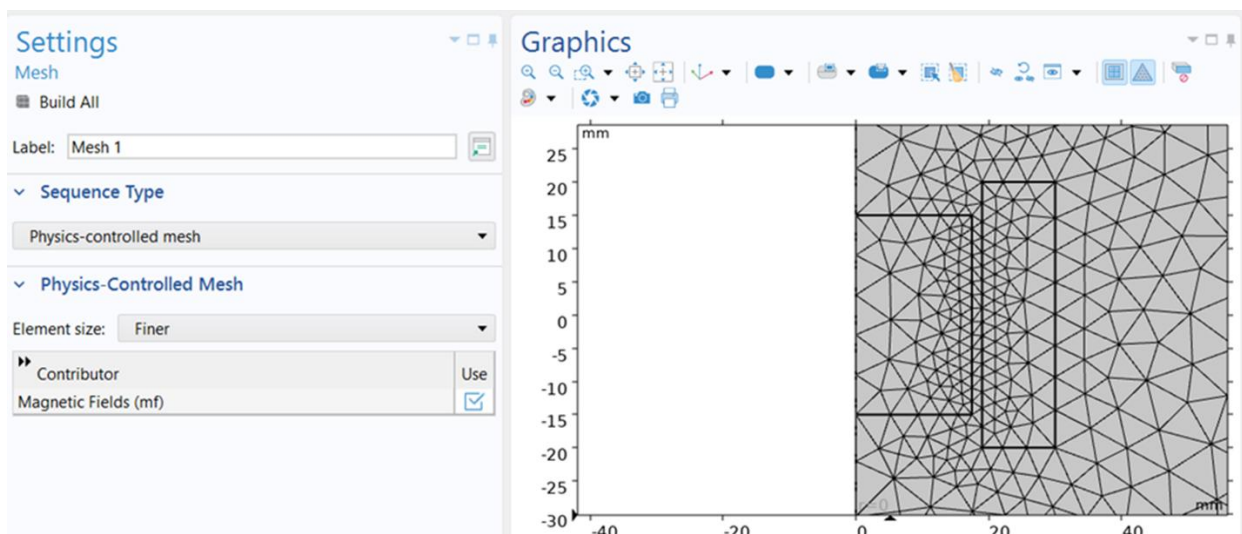


Figure 3.23: Mesh distribution used in the COMSOL simulation

The mesh distribution applied to the simulation domain is shown in Fig. 3.24. A physics-controlled mesh with finer element size was used in the magnet and coil region to improve solution accuracy in areas of high magnetic field gradient.

After the practical design was updated based on component availability and prototype requirements, the COMSOL geometry was also revised. The updated simulation geometry included the selected ferrite ring magnets, shaft region, copper coil region, surrounding air domain, and the additional non-magnetic spacer or bush regions used in the moving assembly. This revised geometry was developed to better represent the actual prototype configuration used for fabrication and testing.

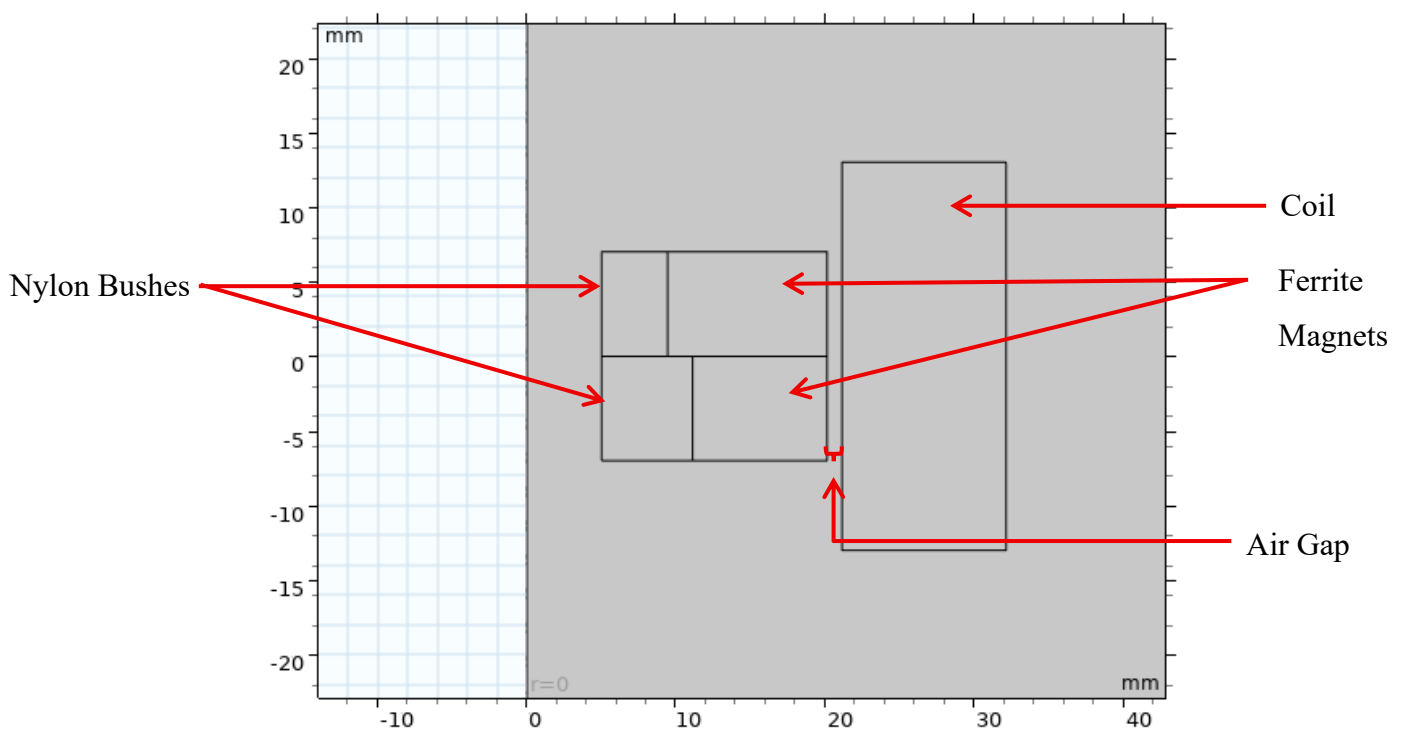


Figure 3.24: Updated 2D axisymmetric geometry of the electromagnetic vibration energy harvester used in COMSOL simulation

The updated geometry was based on the practical moving assembly used in the prototype. The ferrite magnets were modelled according to the selected commercial dimensions, while the non-magnetic bush regions were included to represent the actual spacing and mass-adjustment arrangement. This allowed the simulation to more closely represent the real physical system

3.9.3 Material Properties

The material properties were assigned based on the actual prototype components. In the initial simulation stage, an NdFeB-type magnet material was used. However, after commercially available ferrite ring magnets were selected and tested using a gauss meter, the magnet material in COMSOL was updated to BMHFa-30/31 ferrite anisotropic magnet. This update was made to ensure that the simulation model represented the practical prototype more accurately.

The materials used in the simulation are summarized in Table 3.11.

Table 3.11: Materials used in simulations

Component	Material
Ring type cylindrical magnet	BMHFa-30/31 ferrite anisotropic magnet
Coil	Copper
Harvester casing	PLA
Nylon bush	Nylon / non-magnetic polymer material
Core	Low Carbon Steel 1010
Surrounding domain	Air

The nylon bushes were included in the design as non-magnetic spacer components. Their main function was mechanical rather than electromagnetic. Therefore, they were treated as non-magnetic polymer regions in the simulation, like the casing material.

The use of a non-magnetic casing material was important because the casing should provide mechanical support without significantly affecting the magnetic field. Preliminary comparison between casing and no-casing arrangements showed that the casing had negligible influence on magnetic flux linkage. Therefore, later simulations were simplified by excluding the casing from the electromagnetic model while retaining PLA as the intended prototype housing material.

3.9.4 Boundary Conditions and Physics Setup

The electromagnetic simulation was carried out using the magnetic field physics interface in COMSOL Multiphysics. The model was developed to evaluate how the magnetic field produced by the permanent magnet links with the copper coil and how this flux linkage changes when the magnet moves relative to the coil.

The surrounding air domain was used to represent the open space around the electromagnetic components. External magnetic insulation or suitable outer boundary conditions were applied at the outer boundary of the air domain to limit the computational region while still allowing the magnetic field to be solved properly inside the model.

For the parametric sweep study, the magnet position was varied relative to the coil. This allowed the magnetic flux linkage through the coil to be calculated at different magnet positions. For the time-dependent study, the magnet motion was defined as a sinusoidal displacement input. This created a changing magnetic flux linkage with time, producing an induced voltage in the coil.

The induced voltage was evaluated based on the rate of change of magnetic flux linkage. According to Faraday's law, voltage is induced when the magnetic flux linking the coil changes with time. Therefore, the time-dependent simulation was necessary to estimate the voltage waveform produced by vibration-type motion.

3.9.5 Simulation Cases

Several simulation cases were considered to evaluate the influence of different design features on electromagnetic performance. The main purpose of these cases was to identify whether certain components significantly improved flux linkage or whether they could be removed to simplify the design.

The main simulation cases are summarized in Table 3.12.

Table 3.12: COMSOL simulation cases

Case	Simulation condition	Purpose
Case 1	Harvester with non-magnetic casing	To evaluate whether the casing affects magnetic flux distribution
Case 2	Harvester without casing	To check whether the model can be simplified without affecting results
Case 3	Harvester with ferromagnetic core	To evaluate whether the core improves flux linkage
Case 4	Core-less harvester	To evaluate the effect of removing the core and allowing greater relative motion
Case 5	Parametric magnet displacement sweep	To obtain flux linkage variation with magnet position
Case 6	Time-dependent sinusoidal vibration input	To obtain induced voltage response with time

The key parameters used in the COMSOL simulation are summarized in Table 3.13, including the simulation type, selected settings, and their purpose in reducing model complexity and improving computational efficiency.

Table 3.13: Summary of key COMSOL simulation parameters

Parameter	Value / Setting	Purpose
Simulation type	2D axisymmetric	Reduce computational time while representing cylindrical geometry
Physics interface	Magnetic Fields	Analyse magnetic flux and flux linkage
Magnet displacement range	-20 mm to +20 mm	Study flux linkage variation with position

Displacement step size	1 mm	Capture smooth flux linkage trend
Time-dependent input	Sinusoidal displacement	Represent vibration-type motion
Assumed amplitude	20 mm / 2 cm	Preliminary voltage simulation input
Assumed frequency	10 Hz	Preliminary voltage simulation input
Casing material	PLA	Check effect of non-magnetic housing
Coil material	Copper	Represent conducting coil region
Magnet material	BMN-35 / NdFeB-type magnet	Represent permanent magnet field

The casing comparison was carried out because the housing is required for prototype fabrication, but it should not strongly affect the magnetic field. Since the casing material is non-magnetic, it was expected to have minimal influence on the magnetic flux distribution.

The core and core-less comparison was performed to determine whether a ferromagnetic core was beneficial for the selected geometry. Although a core can concentrate magnetic flux, it can also reduce available space and mechanically restrict relative motion. Therefore, both configurations were compared using flux linkage results.

The time-dependent voltage simulation was then performed using an assumed vibration amplitude of 2 cm and an assumed vibration frequency of 10 Hz. These values were used for preliminary simulation and were intended to be updated later using experimentally measured vehicle vibration data.

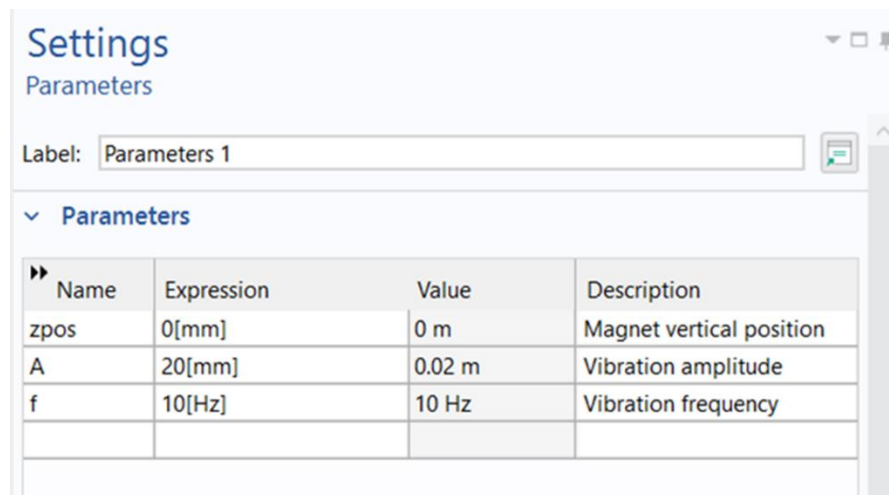


Figure 3.25: Parametric sweep settings used for magnet displacement in COMSOL

The parametric sweep settings used for the displacement study are shown in Fig. 3.26. The magnet position was varied relative to the coil to calculate the corresponding change in flux linkage. This allowed the relationship between magnet displacement and coil flux linkage to be evaluated before conducting the time-dependent voltage simulation.

3.9.6 Magnetic Flux Distribution

Magnetic flux distribution was analysed to observe how the magnetic field passes through the coil region and surrounding air domain. This helped to confirm whether the magnet and coil were positioned correctly and whether the selected geometry produced useful magnetic field interaction with the coil.

The flux density plots showed the magnetic field concentration around the permanent magnet and the flux path through the nearby regions. In the casing comparison, the flux distribution with the non-magnetic casing was very similar to the configuration without casing. This confirmed that the casing did not significantly disturb the magnetic field.

The magnetic flux density distribution with casing is shown in Fig. 3.27.

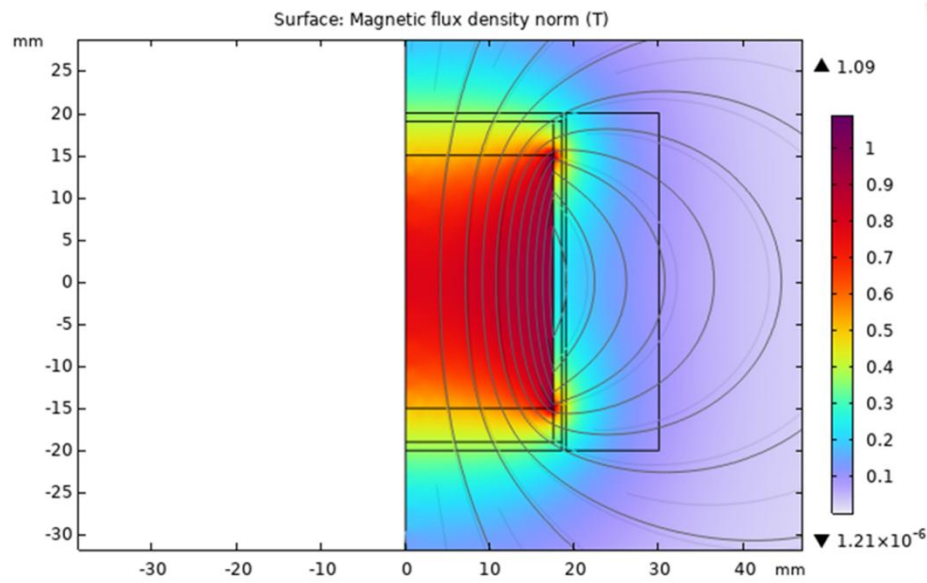


Figure 3.26: Magnetic flux density distribution with casing

High flux density (up to ~ 1.09 T) is concentrated in the magnet region, with field lines extending symmetrically into the surrounding air domain.

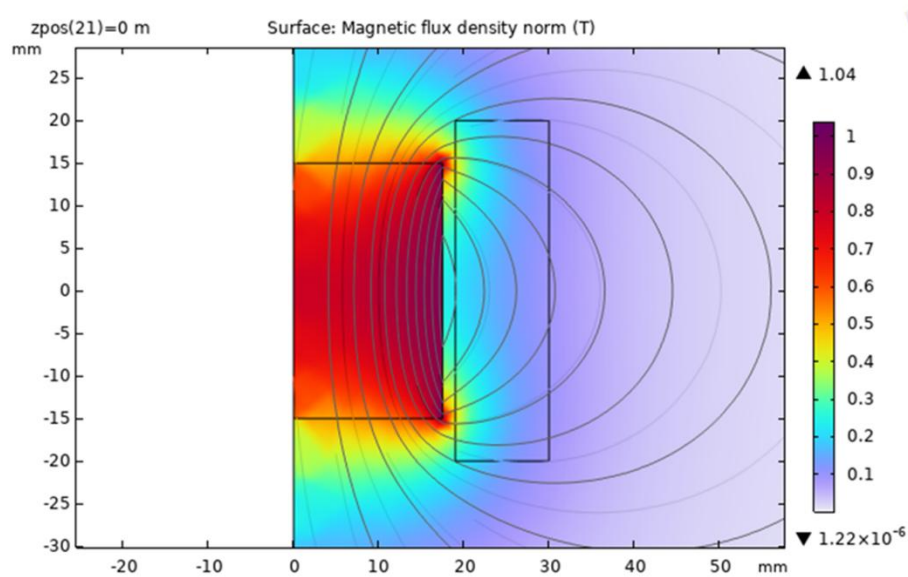


Figure 3.27: Magnetic flux density distribution without casing

The similarity between Fig. 3.27 and Fig. 3.28 confirms that the non-magnetic casing does not significantly affect the magnetic field pattern.

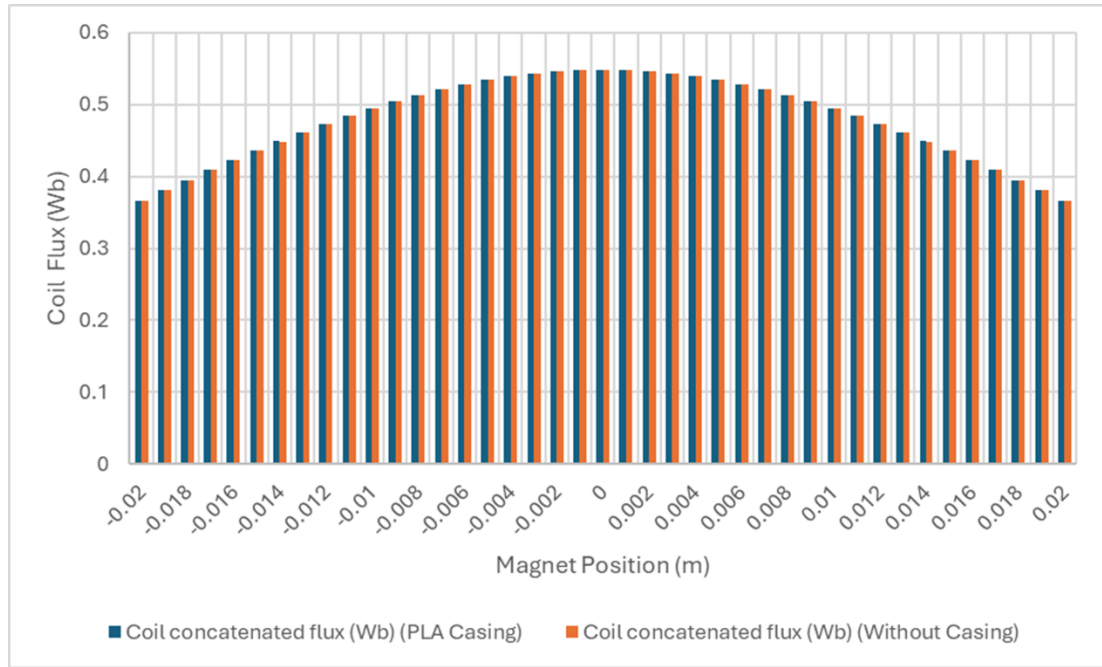


Figure 3.28: Flux linkage comparison between casing and no-casing configurations

The flux linkage curves are nearly identical, confirming that the PLA casing can be excluded from later electromagnetic simulations without significantly affecting the result.

The comparison between the casing and no-casing configurations showed that the PLA casing had negligible influence on the magnetic flux distribution and flux linkage. This is because PLA is a non-magnetic material and does not significantly disturb the magnetic field around the magnet and coil. Therefore, the casing was retained for prototype fabrication but removed from later electromagnetic simulations to simplify the model and reduce computational time.

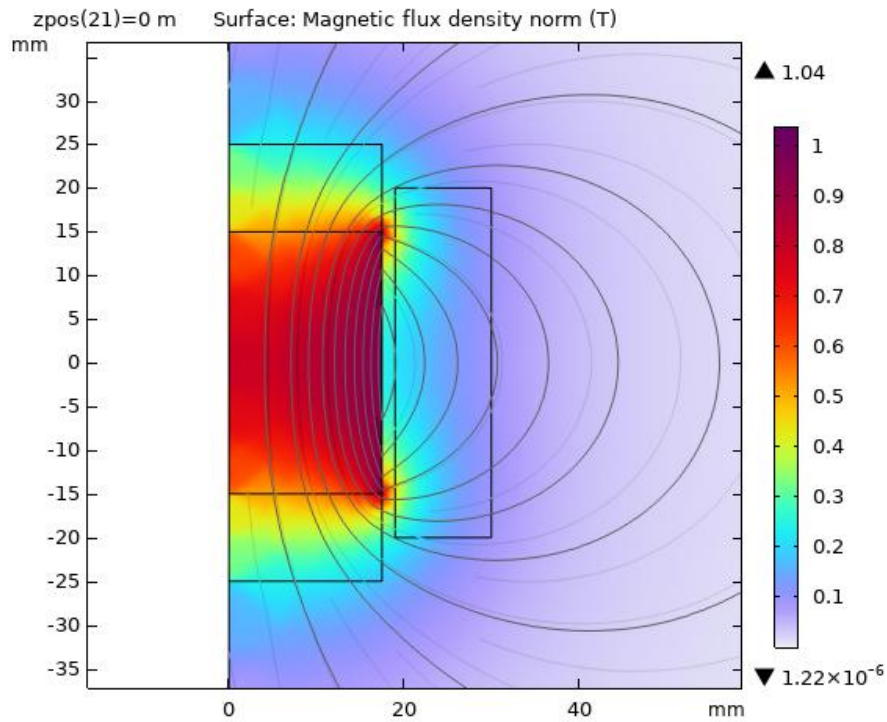


Figure 3.29: Magnetic flux density distribution with ferromagnetic core

After updating the harvester geometry and material properties, the magnetic flux distribution was re-evaluated using the revised COMSOL model. The updated magnetic flux density plot is shown in Fig. 3.31. This result was used to confirm the magnetic field concentration around the ferrite magnet region and to observe how the magnetic field links with the copper coil in the updated configuration.

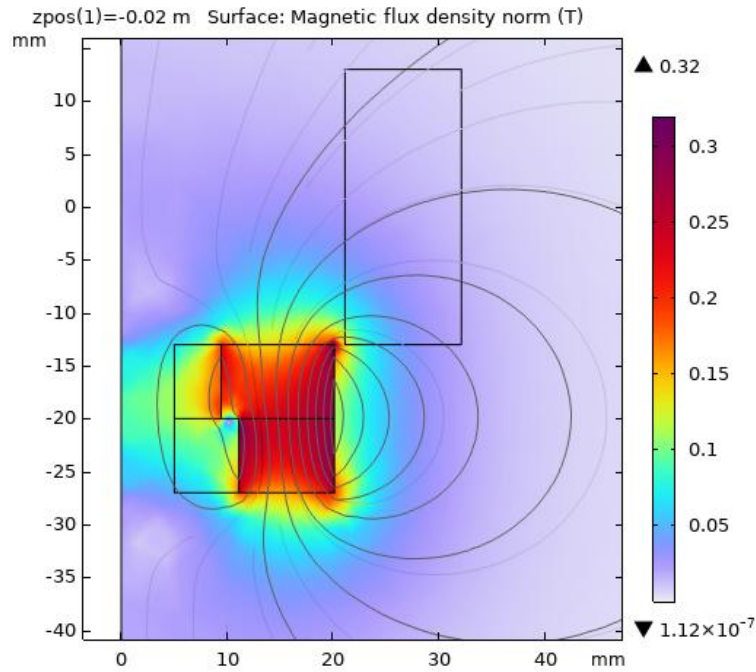


Figure 3.30: Magnetic flux density distribution of the updated electromagnetic harvester configuration

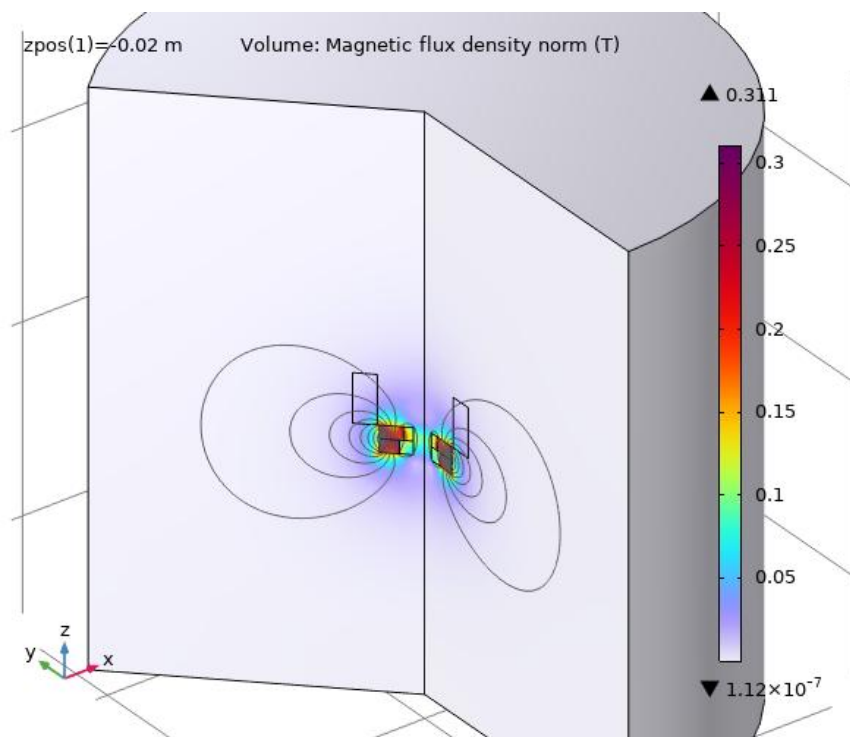


Figure 3.31: 3D magnetic flux density distribution of the updated electromagnetic vibration energy harvester model

The updated flux density distribution shows that the magnetic field is concentrated mainly around the ring-type magnet region and extends through the nearby coil region. This confirms

that the selected ferrite magnets can produce a useful magnetic field for electromagnetic induction, even though the magnet material was changed from the initially assumed NdFeB type to the practically available ferrite type.

3.9.7 Simulation Limitations

Although the COMSOL simulation provided useful design insight, several limitations should be acknowledged.

First, the time-dependent voltage simulation was initially performed using assumed vibration input values. The vibration amplitude was assumed as 2 cm, and the excitation frequency was assumed as 10 Hz. Therefore, the voltage output does not yet represent the final real-world vehicle vibration response. The simulation must be updated using experimentally measured vibration data for more accurate prediction.

Second, the simulation mainly focused on electromagnetic behavior. Mechanical losses such as friction in the shaft, spring damping, impact between components, misalignment, and manufacturing tolerances were not fully included. These effects can reduce the actual relative motion and therefore reduce the generated voltage in the physical prototype.

Third, the electrical output was mainly evaluated at the coil level. Practical power conversion losses from rectification, voltage regulation, battery charging, and power management circuitry were not fully included in the COMSOL model. Therefore, the simulated induced voltage cannot be directly used to claim that the GPS tracker can be powered without further system-level testing.

Fourth, the model used ideal or simplified material properties. In the actual prototype, magnet strength, coil resistance, winding quality, casing accuracy, and assembly tolerances can differ from the simulation assumptions.

Finally, the simulation did not represent all possible vehicle vibration conditions. Real vehicle vibrations are broadband, irregular, and depend on road condition, speed, mounting position, and vehicle structure. Therefore, future simulations should use measured vibration profiles as input instead of a single sinusoidal excitation.

These limitations do not reduce the value of the simulation, but they show that COMSOL results should be used as a design guide rather than final proof of system performance.

3.10 Fabricated Prototype for the Mechanical Model

Using the parameter calculation from our analytical model, a 3D model was designed for the EM energy harvester using SOLIDWORKS.

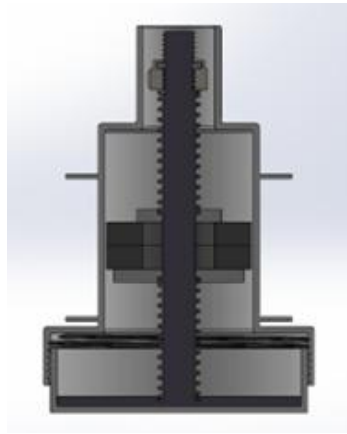


Figure 3.32: Section view of the 3D model

The detailed exploded views of the 3D model and the production drawings of the individual components are provided in the Appendix. These drawings were used as the basis for manufacturing and assembling the physical mechanical prototype. The fabricated model was therefore developed directly from the finalized CAD design and its corresponding engineering drawings ensuring consistency between the analytical design, digital model and physical prototype.



Figure 3.33: Fabricated mechanical model

The fabricated mechanical model of the vibration energy harvester is shown in Figure 3.33. The prototype consists of a central guide shaft, spring-supported moving mass assembly end

supports, and fastening components arranged to produce vertical oscillatory motion. The mechanical parameters of this model were not selected arbitrarily. They were determined using the analytical mathematical model developed for the harvester. The target natural frequency was selected from the vehicle vibration analysis and the required proof mass and spring stiffness were calculated using the mass-spring relationship $K = M(2\pi f_0)^2$. The spring arrangement was then selected to provide the required equivalent stiffness while the moving mass was designed to provide sufficient inertia for resonance based vibration amplification. This fabricated model therefore represents the physical implementation of the analytical design allowing the predicted resonance behaviour and relative motion of the harvester to be tested experimentally.

4 Analysis and Results

4.1 Simulation Results

4.1.1 Flux Linkage Variation

Flux linkage variation was analysed because induced voltage depends on how rapidly the magnetic flux linkage through the coil changes with time. A stationary parametric sweep was carried out by changing the magnet position relative to the coil. This allowed the coil flux linkage to be calculated at each displacement position.

The magnet position varied from -20 mm to +20 mm in 1 mm steps. The flux linkage increased as the magnet moved toward the central coil region and reached its maximum value near the central position. After passing the centre, the flux linkage reduced symmetrically. This symmetric trend indicates that the geometry and magnet-coil alignment were modelled correctly.

Table 4.1: Sample flux linkage comparison between core and core-less configurations

Magnet position (m)	Without core, coil concatenated flux (Wb)	With core, coil concatenated flux (Wb)
-0.020	0.36674	0.36675
-0.010	0.49454	0.49455
0.000	0.54824	0.54826
+0.010	0.49454	0.49456
+0.020	0.36674	0.36676

The full flux linkage data from the parametric displacement sweep is provided in Appendix.

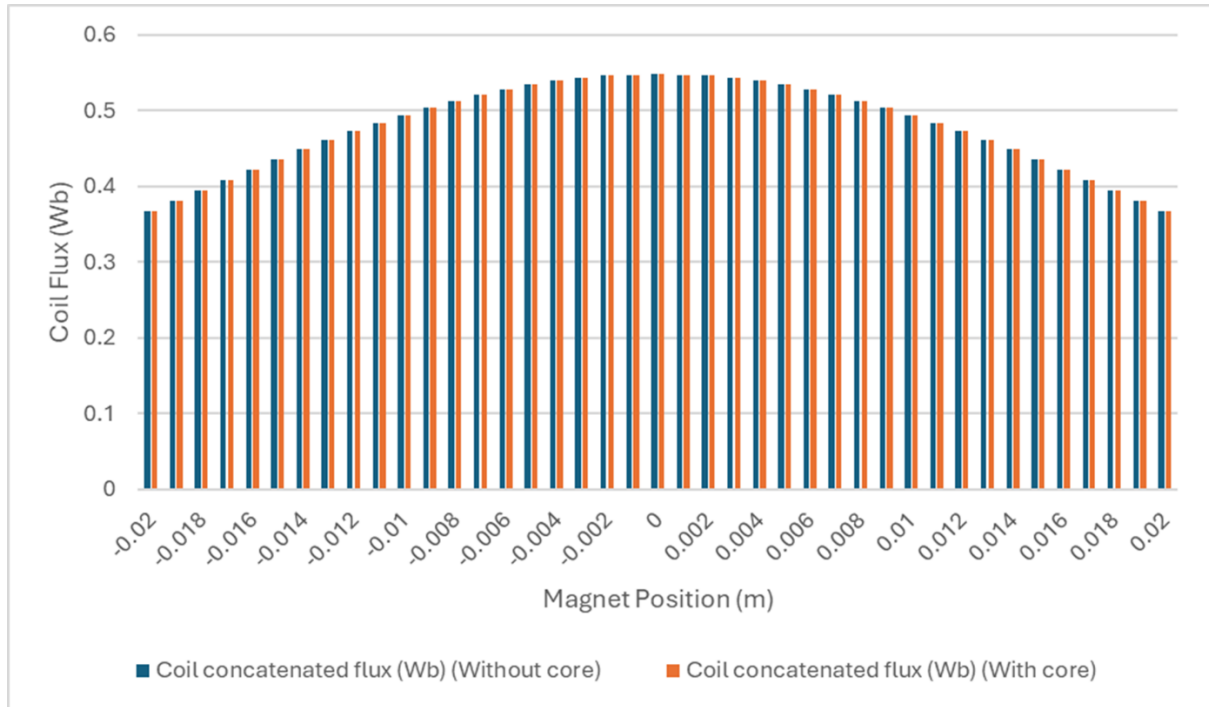


Figure 4.1: Flux linkage variation with magnet displacement for core and core-less configurations

Table 4.1 shows selected flux linkage values from the displacement sweep. The maximum flux linkage occurred at the central magnet position, where the value was 0.54824 Wb without the core and 0.54826 Wb with the core. The difference between the two configurations was very small, indicating that the ferromagnetic core did not significantly improve flux linkage for the selected geometry. Although the core helped guide the magnetic field locally, it also increased design complexity and could restrict the available space for magnet movement. Therefore, the core-less configuration was selected for further development because it provides a simpler mechanical arrangement while maintaining almost the same flux linkage performance.

Table 4.2: Selected coil concatenated flux values at different magnet z-positions for the updated harvester configuration

Magnet position (m)	Coil concatenated flux (Wb)
-0.020	0.047535
-0.015	0.065810
-0.010	0.084677
-0.005	0.098239
0.000	0.102590
0.005	0.097648
0.010	0.083628
0.015	0.064580
0.020	0.046409

The simulation results show that the coil concatenated flux increased as the magnet moved toward the centre of the coil region. The maximum flux value of approximately 0.10259 Wb occurred at middle position (0 m), which corresponds to the central aligned position between the magnet and the coil. After passing the centre, the flux decreased gradually as the magnet moved away from the coil region.

The variation is approximately symmetric about the central position. This indicates that the geometry, magnet alignment, and coil placement in the updated harvester model were correctly defined. The result also confirms that the selected configuration can produce significant flux linkage variation with displacement, which is necessary for induced voltage generation.

4.1.2 Induced Voltage Simulation

A time-dependent simulation was carried out to estimate the induced voltage generated in the coil due to vibration-type motion. In this simulation, the relative motion between the magnet and coil was represented using a sinusoidal displacement input. The initial simulation used an assumed vibration amplitude of 2 cm and a frequency of 10 Hz. These values were used to establish the simulation workflow and will be replaced with measured vehicle vibration data during final validation.

The induced voltage was generated because the magnet motion caused the magnetic flux linkage through the coil to vary with time. As the magnet moved upward and downward relative to the coil, the flux linkage changed continuously. This produced an alternating voltage waveform at the coil terminals.

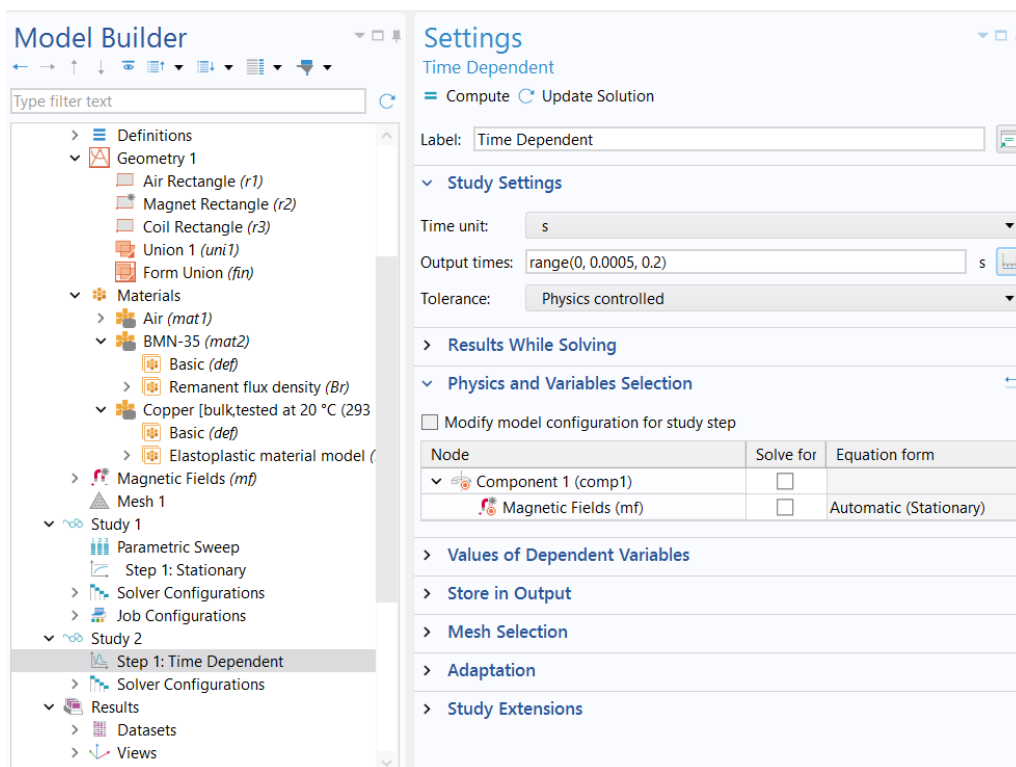


Figure 4.2: Time-dependent simulation settings used for sinusoidal vibration input

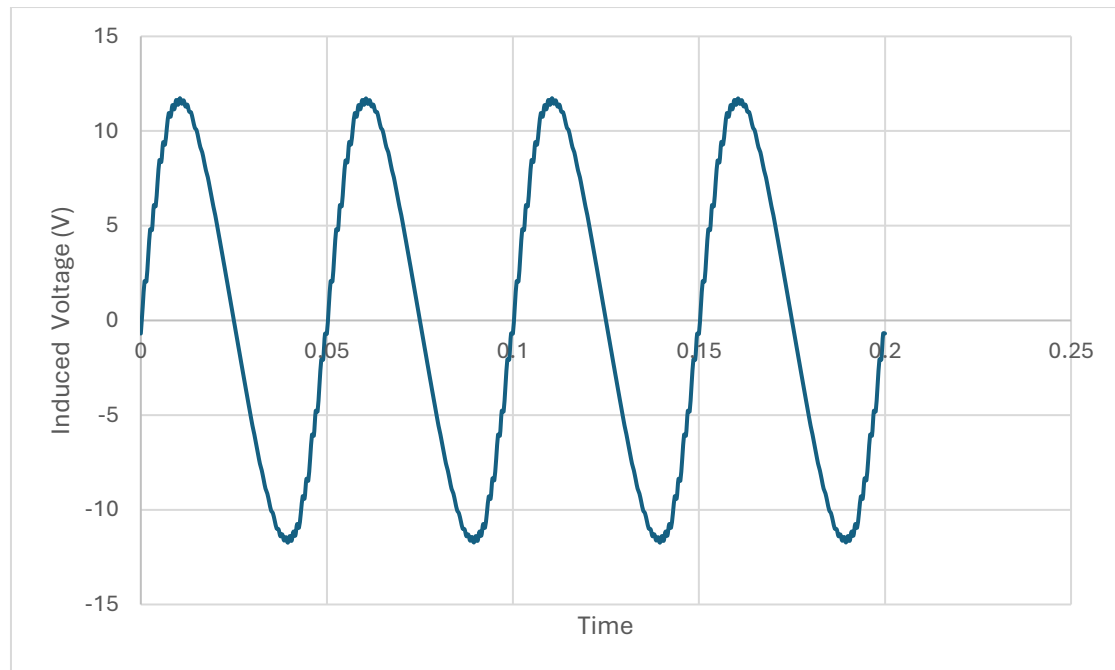


Figure 4.3: Induced voltage waveform obtained from the time-dependent COMSOL simulation

The induced voltage waveform in Fig. 4.3 shows that the selected magnet–coil arrangement can generate an alternating voltage response when subjected to sinusoidal vibration input. The voltage is produced because the relative motion between the magnet and coil changes the magnetic flux linkage with time. However, this result should be treated as a preliminary prediction because the simulation used assumed vibration amplitude and frequency values. Final validation should use experimentally measured vehicle vibration data as the excitation input.

Table 4.3: Sample induced voltage values obtained from the time-dependent simulation of the updated harvester configuration.

Time (s)	Induced voltage (V)	Time (s)	Induced voltage (V)
0.0000	-0.69115	0.1000	-0.69115
0.0050	8.46150	0.1105	11.73700
0.0105	11.73700	0.1250	0.00000
0.0250	0.00000	0.1395	-11.73700

0.0395	-11.73700	0.1500	-0.70372
0.0500	-0.70372	0.1605	11.72700
0.0605	11.72700	0.1750	0.00000
0.0750	0.00000	0.1895	-11.72700
0.0895	-11.72700	0.2000	-0.69115

The maximum positive induced voltage was approximately 11.737 V, while the maximum negative induced voltage was approximately -11.737 V. The waveform crosses near zero at regular intervals, confirming the alternating nature of the induced voltage. This shows that the updated ferrite magnet and coil arrangement can generate voltage under the assumed sinusoidal vibration input. However, this result should be treated as a preliminary simulation prediction because the excitation input was assumed. Practical output after rectification, voltage regulation, battery charging, and power management losses may be lower than the simulated coil voltage.

4.1.3 Key Simulation Findings

The COMSOL simulation provided several important findings for the design and development of the electromagnetic vibration energy harvester.

The main findings are summarized in Table 4.4.

Table 4.4: Simulation main findings

Finding	Explanation
2D axisymmetric modelling was suitable	The cylindrical geometry allowed the model to be simplified while maintaining the main electromagnetic behavior.
Non-magnetic casing had negligible effect	The casing did not significantly change the magnetic flux linkage, so it could be removed from later simulation models.

Flux linkage varied symmetrically	The symmetric variation confirmed correct magnet-coil alignment in the model.
Core effect was very small	The difference between core and core-less flux linkage was minimal for the selected geometry.
Core-less design is more practical	It reduces complexity and allows greater relative motion of the moving magnet assembly.
Induced voltage was generated under sinusoidal input	The time-dependent simulation confirmed that vibration-type motion can produce voltage in the coil.
Further optimization is required	Parameters such as coil turns, magnet size, air gap, and vibration input should be optimized for better output.

Overall, the simulation results support the feasibility of the selected linear electromagnetic harvester concept. The model showed stable magnetic flux behavior, measurable flux linkage variation, and induced voltage generation under assumed vibration input. The simulation framework is also suitable for further parametric studies such as air-gap optimization, coil geometry optimization, and magnet size selection.

5 Conclusion and Futureworks

5.1 Conclusion

The overall aim of this project, as set out in the introduction, was to develop a small scale vibration based energy harvester that can serve as an alternative power source for low power portable sensors, in particular for a GPS tracking unit which currently depends on a primary battery and therefore requires regular replacement. To address this need, the work has progressed through a structured engineering design process which combines a literature review, analytical modelling, numerical simulation and physical fabrication, with each stage informing the next.

Drawing the previous chapters together, the project has shown that the kinetic energy available in typical low frequency ambient vibrations can be converted into useful electrical energy using a cantilever based harvester whose mechanical parameters are deliberately matched to the dominant excitation frequency of the intended host. The literature review highlighted a clear gap between the high frequency, laboratory scale devices that are commonly reported in the existing publications and the relatively low frequency, broadband vibration profiles that are actually present in mobile applications such as vehicles and railway assets. The analytical calculations presented in the methodology chapter were used to estimate the natural frequency, the displacement amplitude and the expected order of magnitude of the electrical output, and the finite element simulation was then used to verify these analytical predictions and to refine the geometry of the cantilever and the proof mass before any material was committed. Finally, the fabrication stage has confirmed that the design can be realised with the workshop level facilities available to the group, and that the assembled prototype responds, at least under preliminary manual excitation, in a manner consistent with the predicted vibration mode shape.

Bringing these strands together, the central conclusion of this study is therefore not simply that a working harvester prototype has been produced, but more importantly that a repeatable design route has been demonstrated, in which the target frequency band is first identified from the application, then translated into a set of mechanical parameters through analytical and numerical models, and finally embodied in a manufacturable assembly. In this way, the project has narrowed the gap which was identified in the introduction between the generic harvester models found in the literature and the specific power requirement of a custom built, self powered GPS tracker.

At the same time, it is necessary to be self critical about what the present study has, and has not, established. The most important limitation is that the experimental validation of the electrical output has not been completed within the available project duration, which means that the predicted voltage and power levels still remain unverified, and the analytical and finite element models have not yet been calibrated against real measurements. The modelling itself was based on a single degree of freedom, lumped parameter representation and on the assumption of a purely sinusoidal input, which is unlikely to fully capture the broadband and non-stationary nature of real vibration sources, or the geometric and damping non-linearities that may appear at larger displacement amplitudes. In addition, small geometric deviations and clamping inconsistencies which were observed during the assembly stage are expected to shift the resonant frequency of the physical prototype slightly away from the simulated value. Taken together, these limitations do not invalidate the design, but they clearly define the boundary of what can reasonably be claimed at the present stage: namely that a candidate prototype now exists, and that its expected performance is supported by theory and simulation but still awaits experimental confirmation.

5.2 Future Work

Based on the limitations discussed in the conclusion, the following directions have been identified as the main areas of future work:

- Carry out the experimental characterization of the fabricated prototype by mounting it on a vibration shaker table and exciting it at a series of known and controlled frequencies, while recording the voltage and power output using a calibrated data acquisition system, in order to obtain the actual response curve of the device and to extract the true natural frequency, the damping ratio and the peak power output, which can then be used to calibrate the analytical and finite element models.
- Acquire real vibration data from the intended host and apply signal processing techniques such as the Fast Fourier Transform and the power spectral density analysis to identify the dominant frequency components, so that the mechanical parameters of the harvester, in particular the tip mass and the effective length of the cantilever, can be tuned to bring the natural frequency of the device into alignment with the dominant excitation frequency of the application.

- Design an energy management circuit consisting of a full wave rectification stage, a low loss voltage regulator and a small energy storage element such as a rechargeable lithium polymer cell or a supercapacitor, in order to convert the alternating output of the harvester into a stable DC supply that is suitable for powering digital electronics.
- Develop a custom built GPS tracking unit, comprising a low power GPS receiver, a microcontroller and a wireless communication module, which will be powered directly from the energy stored by the harvester, and field test the complete system in the target environment in order to verify whether the harvested energy is sufficient to sustain the duty cycle that is required for continuous, real time location tracking of the host object.
- Investigate the long term reliability and the broadband performance of the harvester, including the fatigue behaviour of the cantilever element under sustained vibration, the influence of environmental conditions such as temperature, humidity and dust, and the use of non-linear or bi-stable mechanisms or an array of devices with slightly offset natural frequencies, in order to widen the effective operating bandwidth and to move the prototype from a laboratory proof of concept towards a robust engineering product.

6 Reference list

- [1] B. Y. León Ávila, C. A. García Vázquez, O. Pérez Baluja, D. T. Cotfas, and P. A. Cotfas, “Energy harvesting techniques for wireless sensor networks: A systematic literature review,” Jan. 01, 2025, *Elsevier Ltd.* doi: 10.1016/j.esr.2024.101617.
- [2] M. A. A. Abdelkareem *et al.*, “Vibration energy harvesting in automotive suspension system: A detailed review,” Nov. 01, 2018, *Elsevier Ltd.* doi: 10.1016/j.apenergy.2018.08.030.
- [3] J. Eriksson, “Kandidatexamensarbete med inriktning Fordonsteknik Review of Methods for Energy Harvesting from a Vehicle Suspension System,” 2016.
- [4] H. S. Kim, J.-H. Kim, and J. Kim, “A review of piezoelectric energy harvesting based on vibration,” *International Journal of Precision Engineering and Manufacturing*, vol. 12, no. 6, pp. 1129–1141, 2011, doi: 10.1007/s12541-011-0151-3.
- [5] C. Covaci and A. Gontean, “Piezoelectric Energy Harvesting Solutions: A Review,” *Sensors*, vol. 20, no. 12, p. 3512, 2020, doi: 10.3390/s20123512.
- [6] J. Jiang, S. Liu, L. Feng, and D. Zhao, “A Review of Piezoelectric Vibration Energy Harvesting with Magnetic Coupling Based on Different Structural Characteristics,” *Micromachines (Basel)*, vol. 12, no. 4, p. 436, 2021, doi: 10.3390/mi12040436.
- [7] C. Wei and X. Jing, “A comprehensive review on vibration energy harvesting: Modelling and realization,” *Renewable and Sustainable Energy Reviews*, vol. 74, pp. 1–18, 2017, doi: 10.1016/j.rser.2017.01.073.
- [8] M. A. A. Abdelkareem *et al.*, “Vibration energy harvesting in automotive suspension system: A detailed review,” *Appl. Energy*, vol. 229, pp. 672–699, 2018, doi: 10.1016/j.apenergy.2018.08.030.
- [9] C. Fu, J. Lu, W. Ge, C. Tan, and B. Li, “A Review of Electromagnetic Energy Regenerative Suspension System & Key Technologies,” *Computer Modeling in Engineering & Sciences*, vol. 135, no. 3, pp. 1779–1824, 2023, doi: 10.32604/cmes.2022.023092.
- [10] R. Zhang, X. Wang, and S. John, “A Comprehensive Review of the Techniques on Regenerative Shock Absorber Systems,” *Energies (Basel)*, vol. 11, no. 5, p. 1167, 2018, doi: 10.3390/en11051167.
- [11] “How Do GPS Trackers Work: Vehicle GPS Devices Explained.” Accessed: May 15, 2026. [Online]. Available: <https://www.lytx.com/blog/what-is-gps-tracking-and-how-does-it-work>
- [12] “GPS Hardware 101: A Guide to the Components.” Accessed: May 15, 2026. [Online]. Available: <https://www.azuga.com/blog/gps-tracking-hardware>
- [13] “What is the Best GPS Tracker: Types, Pros, and Cons.” Accessed: May 15, 2026. [Online]. Available: <https://www.solidgps.com/best-types-of-gps-trackers/>
- [14] “RAK7200 WisNode Track Lite Datasheet.” Accessed: May 15, 2026. [Online]. Available: <https://docs.rakwireless.com/product-categories/wisnode/rak7200/datasheet/>
- [15] “SIM7000 LTE Shield for Arduino | Botletics — Botletics.” Accessed: May 15, 2026. [Online]. Available: <https://www.botletics.com/products/sim7000-shield>
- [16] “Datasheet_LGT-92”.

Appendices

Appendix 1: Arduino Code for vehicle data acquisition

```
#include <Wire.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_ADXL345_U.h>

#include <SPI.h>
#include <SD.h>

// SD Card SPI Pins
static const int SD_CS = 5;
static const int SD_MOSI = 23;
static const int SD_MISO = 19;
static const int SD_SCK = 18;

// I2C pins
static const int I2C_SDA = 21;
static const int I2C_SCL = 22;

// Logging settings
static const uint16_t SAMPLE_HZ = 100; // 100 Hz logging
static const uint32_t SAMPLE_PERIOD_MS = 1000 / SAMPLE_HZ;
static const uint32_t FLUSH_EVERY_MS = 1000; // flush to SD every 1 second
static const bool PRINT_TO_SERIAL = true; // set false for faster logging

// File name
const char *LOG_PATH = "/vibration.csv";

// SENSOR OBJECT
Adafruit_ADXL345_Unified accel = Adafruit_ADXL345_Unified(12345);

File logFile;

// HELPERS
void die(const char *msg) {
  Serial.println(msg);
  while (true) { delay(100); }
}

void printSensorDetails() {
  sensor_t sensor;
  accel.getSensor(&sensor);
  Serial.println("-----");
  Serial.print ("Sensor: "); Serial.println(sensor.name);
  Serial.print ("Driver Ver: "); Serial.println(sensor.version);
  Serial.print ("Unique ID: "); Serial.println(sensor.sensor_id);
  Serial.print ("Max Value: "); Serial.print(sensor.max_value); Serial.println(" m/s^2");
  Serial.print ("Min Value: "); Serial.print(sensor.min_value); Serial.println(" m/s^2");
  Serial.print ("Resolution: "); Serial.print(sensor.resolution); Serial.println(" m/s^2");
  Serial.println("-----");
}

void setAccelConfig() {
  // Range: choose based on expected vibration/shock
  accel.setRange(ADXL345_RANGE_16_G);

  // Data rate: should be >= your sampling rate
  // If you sample at 100 Hz, set sensor data rate >= 100 Hz (e.g., 200 Hz)
  accel.setDataRate(ADXL345_DATARATE_200_HZ);

  Serial.print("Range set to: +/- ");
  switch (accel.getRange()) {
    case ADXL345_RANGE_16_G: Serial.println("16 g"); break;
    case ADXL345_RANGE_8_G: Serial.println("8 g"); break;
    case ADXL345_RANGE_4_G: Serial.println("4 g"); break;
    case ADXL345_RANGE_2_G: Serial.println("2 g"); break;
    default: Serial.println("? g"); break;
  }

  Serial.print("Data rate set to: ");
  switch (accel.getDataRate()) {
    case ADXL345_DATARATE_3200_HZ: Serial.println("3200 Hz"); break;
```

```

    case ADXL345_DATARATE_1600_HZ: Serial.println("1600 Hz"); break;
    case ADXL345_DATARATE_800_HZ: Serial.println("800 Hz"); break;
    case ADXL345_DATARATE_400_HZ: Serial.println("400 Hz"); break;
    case ADXL345_DATARATE_200_HZ: Serial.println("200 Hz"); break;
    case ADXL345_DATARATE_100_HZ: Serial.println("100 Hz"); break;
    case ADXL345_DATARATE_50_HZ: Serial.println("50 Hz"); break;
    default: Serial.println("other"); break;
  }
}

bool initSD() {
  SPI.begin(SD_SCK, SD_MISO, SD_MOSI);
  if (!SD.begin(SD_CS, SPI, 1000000)) { // 1 MHz SPI
    return false;
  }

  return true;
}

void openLogFile() {
  // Create/append file
  logFile = SD.open(LOG_PATH, FILE_APPEND);
  if (!logFile) die("ERROR: Could not open log file on SD card.");

  // If file is empty, write header
  if (logFile.size() == 0) {
    logFile.println("time_ms,ax_ms2,ay_ms2,az_ms2,ax_g,ay_g,az_g");
    logFile.flush();
  }
}

// ARDUINO SETUP/LOOP
void setup() {
  Serial.begin(115200);
  delay(300);

  Serial.println("\n== ADXL345 Vibration Logger (ESP32 + SD) ==");

  // I2C init
  Wire.begin(I2C_SDA, I2C_SCL);

  // Accelerometer init
  if (!accel.begin()) {
    die("ERROR: No ADXL345 detected. Check wiring + address (0x53/0x1D).");
  }
  printSensorDetails();
  setAccelConfig();

  // SD init
  Serial.println("Initializing SD card...");
  if (!initSD()) {
    die("ERROR: SD initialization failed. Check CS pin, wiring, and SD format (FAT32).");
  }
  Serial.println("SD initialization done.");

  openLogFile();
  Serial.print("Logging to: ");
  Serial.println(LOG_PATH);
  Serial.println("Start logging...");
}

void loop() {
  static uint32_t lastSampleMs = 0;
  static uint32_t lastFlushMs = 0;

  uint32_t now = millis();

  // Sampling scheduler (fixed interval)
  if (now - lastSampleMs >= SAMPLE_PERIOD_MS) {
    lastSampleMs += SAMPLE_PERIOD_MS; // keeps timing steady

    sensors_event_t event;
    accel.getEvent(&event);

    // m/s^2 from library

```

```

float ax = event.acceleration.x;
float ay = event.acceleration.y;
float az = event.acceleration.z;

// Convert to g (1 g = 9.80665 m/s^2)
const float G0 = 9.80665f;
float ax_g = ax / G0;
float ay_g = ay / G0;
float az_g = az / G0;

// CSV line
// time_ms,ax_ms2,ay_ms2,az_ms2,ax_g,ay_g,az_g
logFile.print(now);
logFile.print(",");
logFile.print(ax, 6); logFile.print(",");
logFile.print(ay, 6); logFile.print(",");
logFile.print(az, 6); logFile.print(",");
logFile.print(ax_g, 6); logFile.print(",");
logFile.print(ay_g, 6); logFile.print(",");
logFile.println(az_g, 6);

if (PRINT_TO_SERIAL) {
  Serial.print("t=");
  Serial.print(now);
  Serial.print("ms ax=");
  Serial.print(ax, 3);
  Serial.print(" ay=");
  Serial.print(ay, 3);
  Serial.print(" az=");
  Serial.print(az, 3);
  Serial.println(" (m/s^2)");
}

// Flush periodically to protect data if power is lost
if (now - lastFlushMs >= FLUSH_EVERY_MS) {
  lastFlushMs = now;
  logFile.flush();
}
}

```

Appendix 2: MATLAB codes of the Analytical model

Following MATLAB codes can be found in the link given.

1. MATLAB code for normalized power Vs $k^2 Q_m$
2. MATLAB code for normalized power Vs normalized frequency
3. MATLAB code for normalized power Vs normalized frequency for varying k^2
4. MATLAB code for normalized power Vs normalized frequency for varying Q_m
5. MATLAB code for normalized power Vs normalized frequency for varying ξ_e
6. Sample MATLAB code for PSD Estimation using Welch's Method for Road Induced Vibrations /
7. MATLAB code for Natural Frequency for Harvester Selection /
8. MATLAB code for PLIM Calculation /
9. MATLAB code for Parameter Calculation using COMSOL Simulation's flux density data //
10. MATLAB code for Parameter Calculation using Experimental flux density data using the Gaus Meter //
11. MATLAB code for Power Calculation /
12. MATLAB code for Equivalent Amplitude Calculation

Notes:

For code 6, 7, 8 and 9, accelerometer data from the road test will be needed.

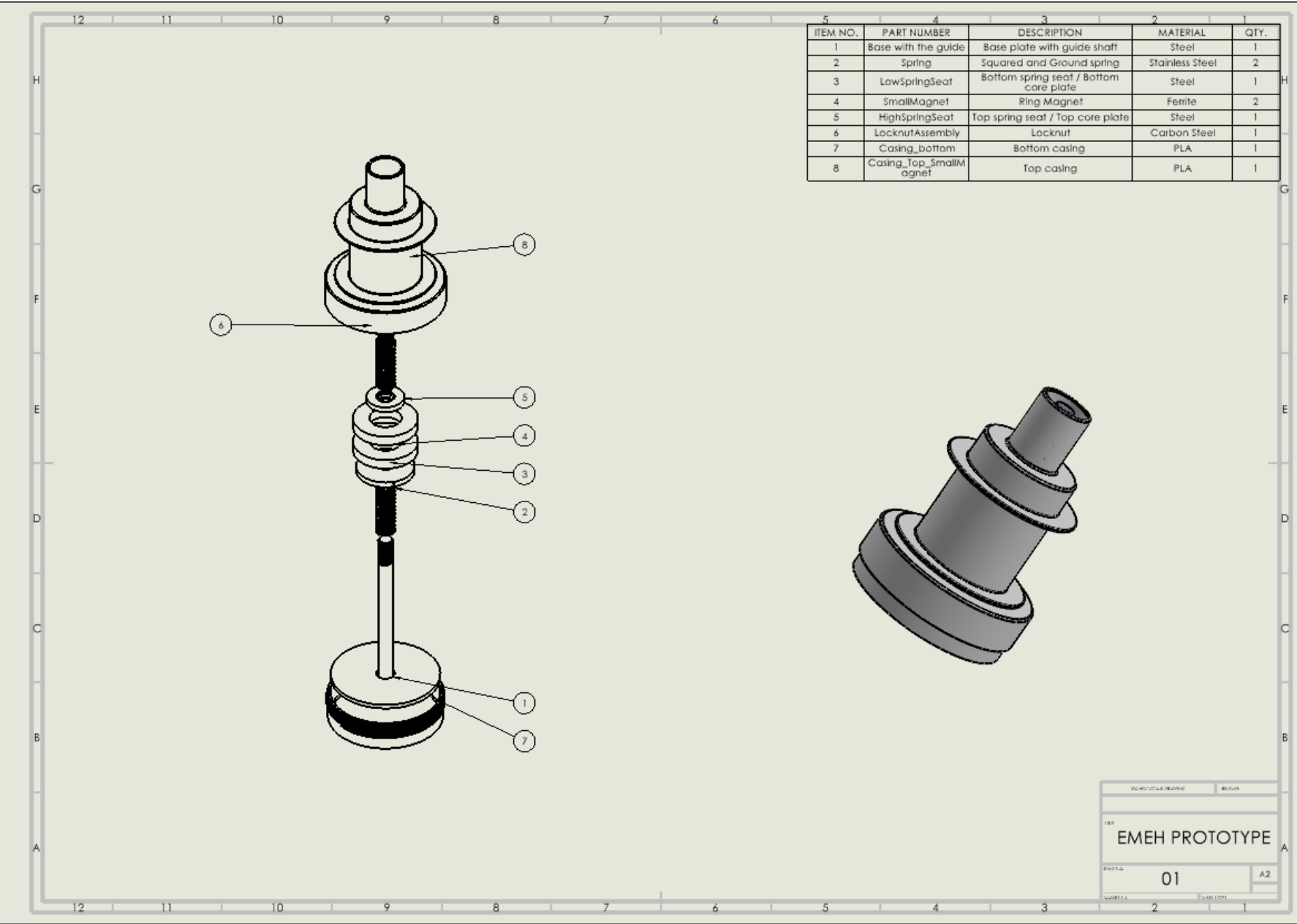
For code 9, COMSOL simulation's flux density data will be needed.

For code 10, magnetic flux density data from the Gaus meter experiment will be needed.

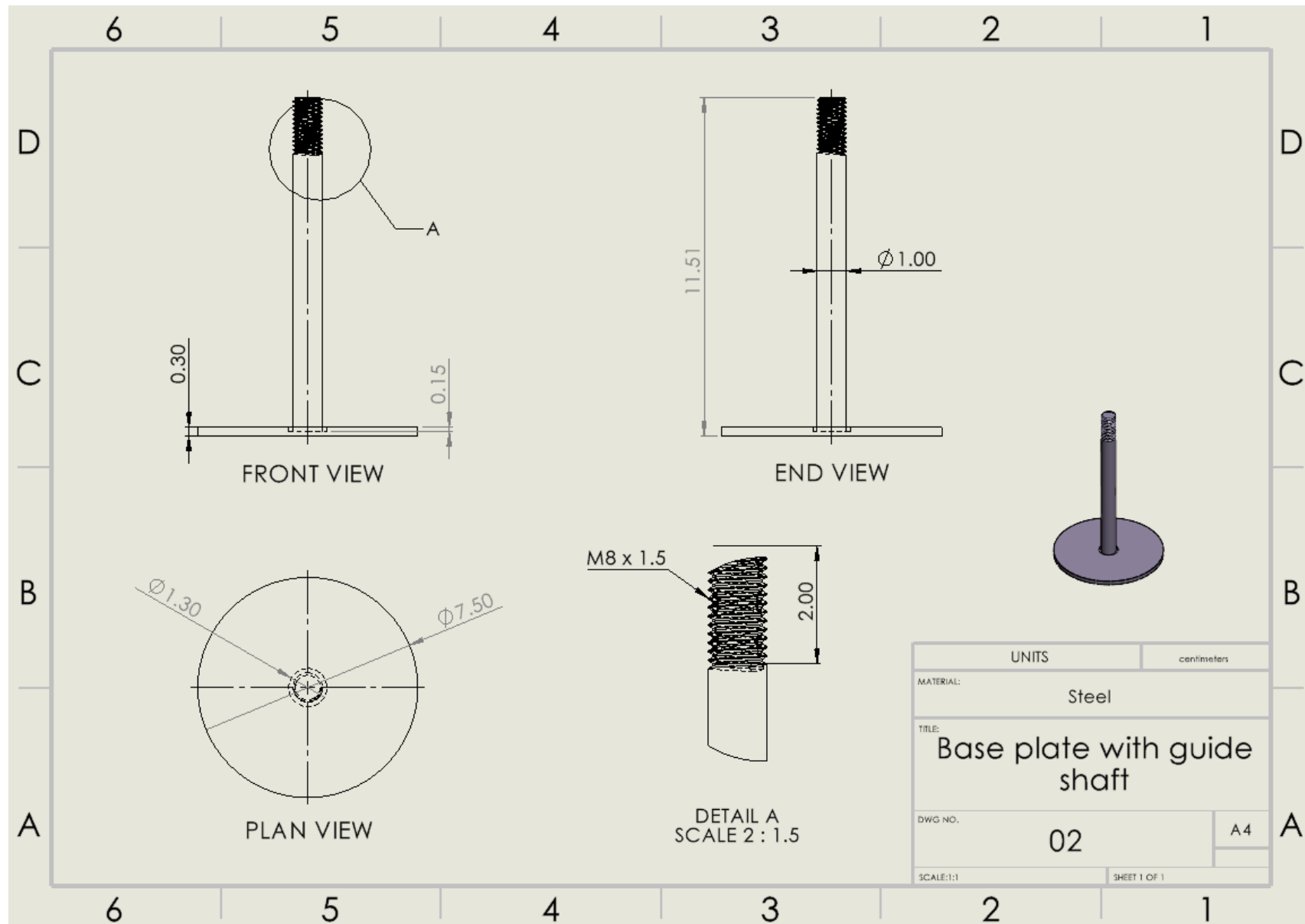
Accelerometer data from the road test, COMSOL simulation's flux density data and magnetic flux density data from the Gaus meter experiment also can be found in the following link.

<https://dms.uom.lk/s/yPfCfPaHdd7ftoK>

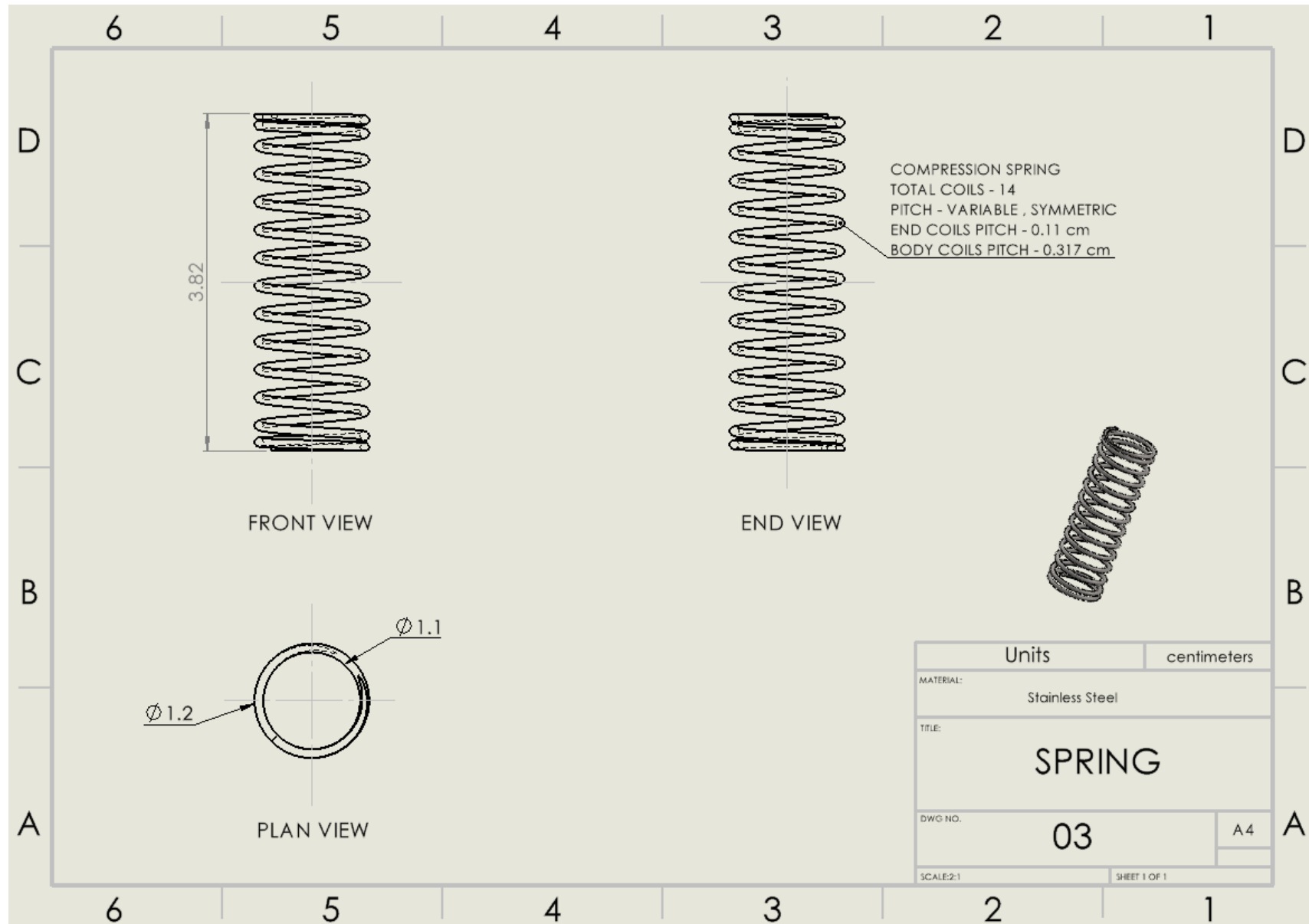
Appendix 3: Exploded View of the EMEH Prototype



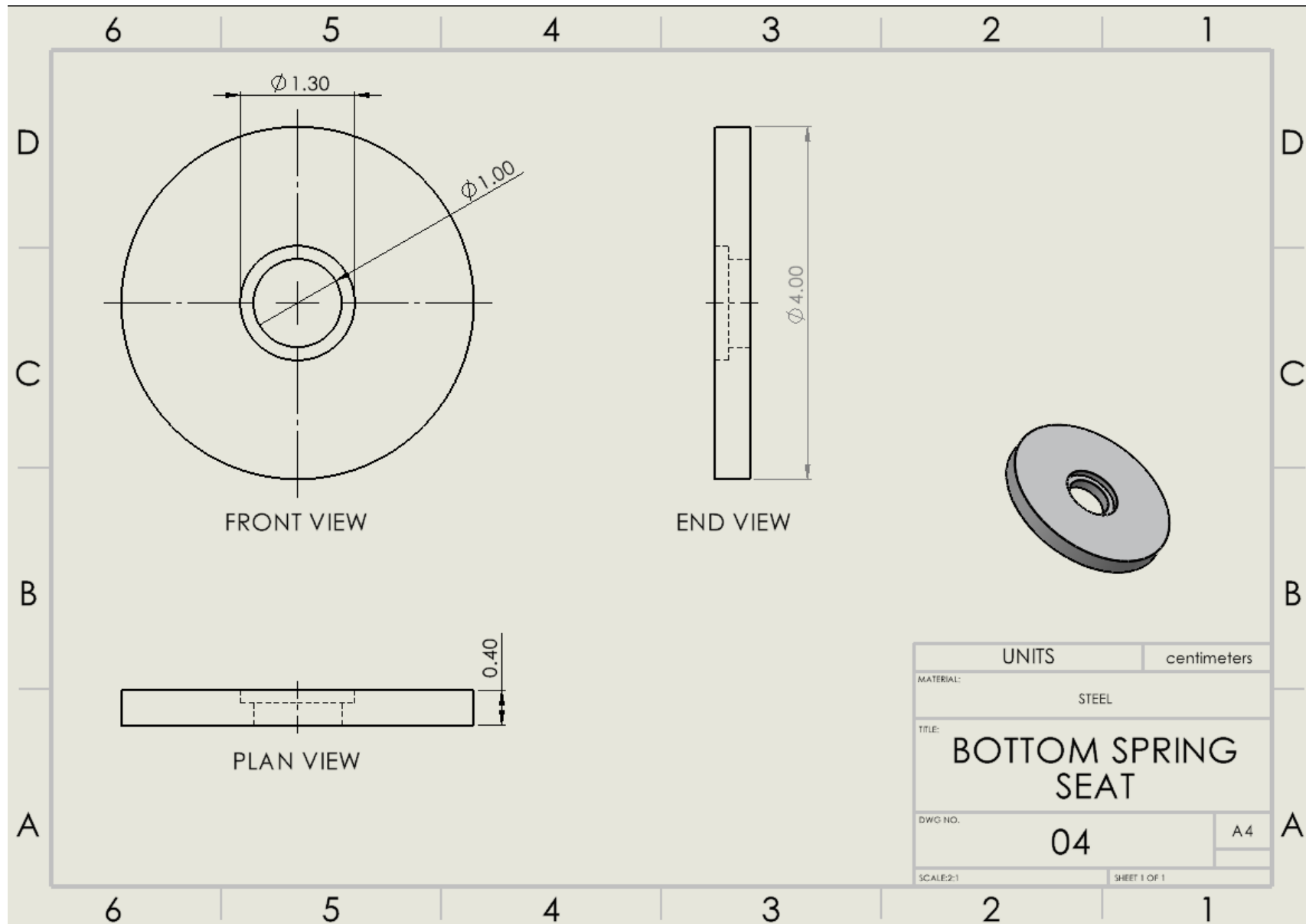
Appendix 4: Production Drawing of base with guide shaft component



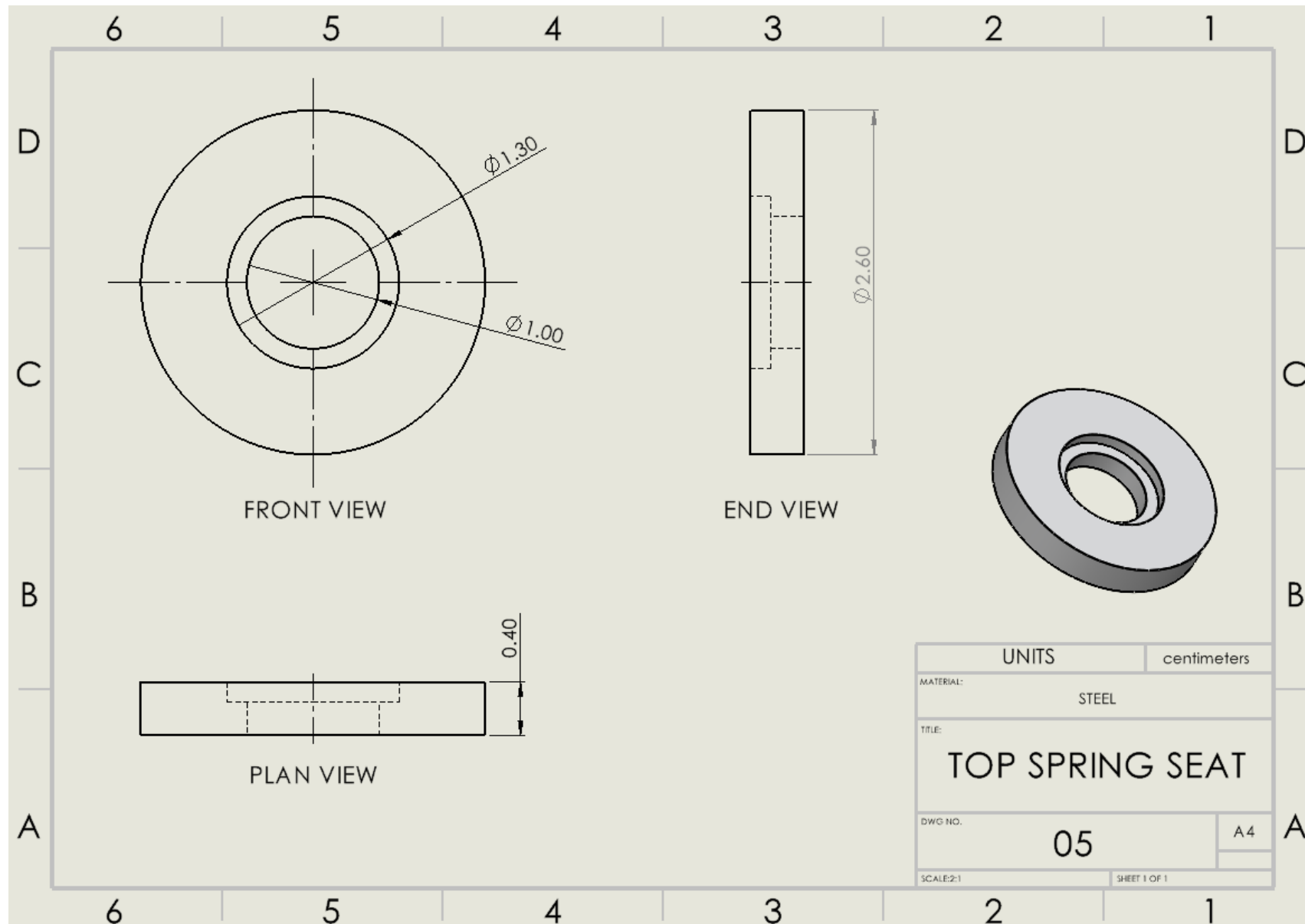
Appendix 5: Production Drawing of Compression Spring component



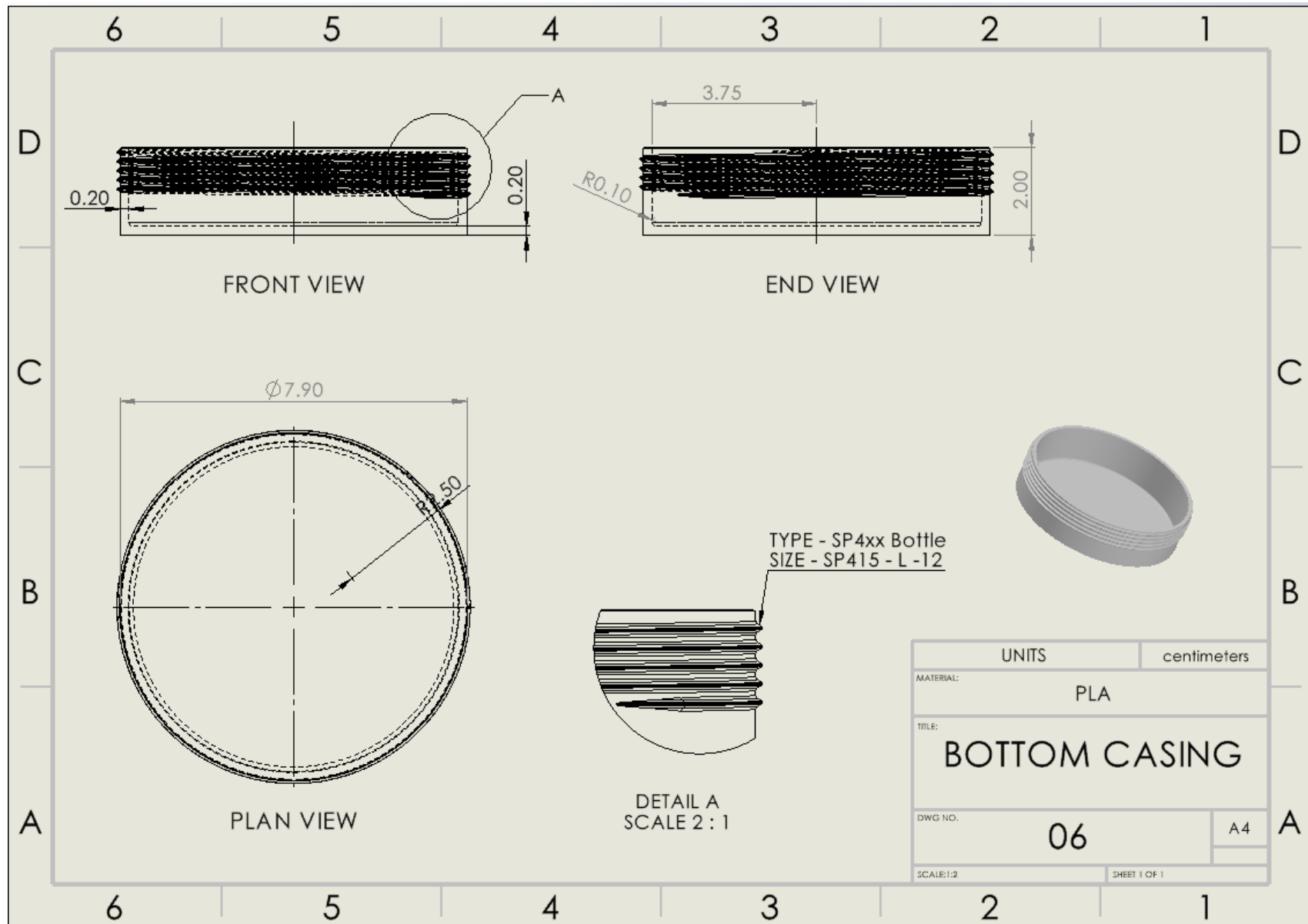
Appendix 6: Production Drawing of Bottom spring seat component



Appendix 7: Production Drawing of Top spring seat component



Appendix 8: Production Drawing of Bottom casing component



Appendix 9: Production Drawing of Top casing component

